

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                        feats_to_balance + feats_grouping +
                        feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	7876
Nombre de lignes df clean	7876
Nombre de features	53
% de NaN	0.0
Taille dataset entraînement	5513
Taille dataset test	2362
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_mean, r_NH3_st_2h_std, r_NH3_st_2h_rms, r_NH3_st_2h_median, r_NH3_st_2h_min, r_NH3_st_2h_max, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_rms, r_NH3_st_8h_median, r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_temperature_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_rms, r_temperature_st_2h_median, r_temperature_st_2h_min, r_temperature_st_2h_max, r_temperature_st_2h_diff_last, r_temperature_st_2h_slope, r_temperature_st_8h_mean, r_temperature_st_8h_std, r_temperature_st_8h_rms, r_temperature_st_8h_median, r_temperature_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_umidity_st_2h_mean, r_umidity_st_2h_std, r_umidity_st_2h_rms, r_umidity_st_2h_median, r_umidity_st_2h_min, r_umidity_st_2h_max,

r_umidity_st_2h_diff_last, r_umidity_st_2h_slope, r_umidity_st_8h_mean, r_umidity_st_8h_std, r_umidity_st_8h_rms, r_umidity_st_8h_median, r_umidity_st_8h_min, r_umidity_st_8h_max, r_umidity_st_8h_diff_last, r_umidity_st_8h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

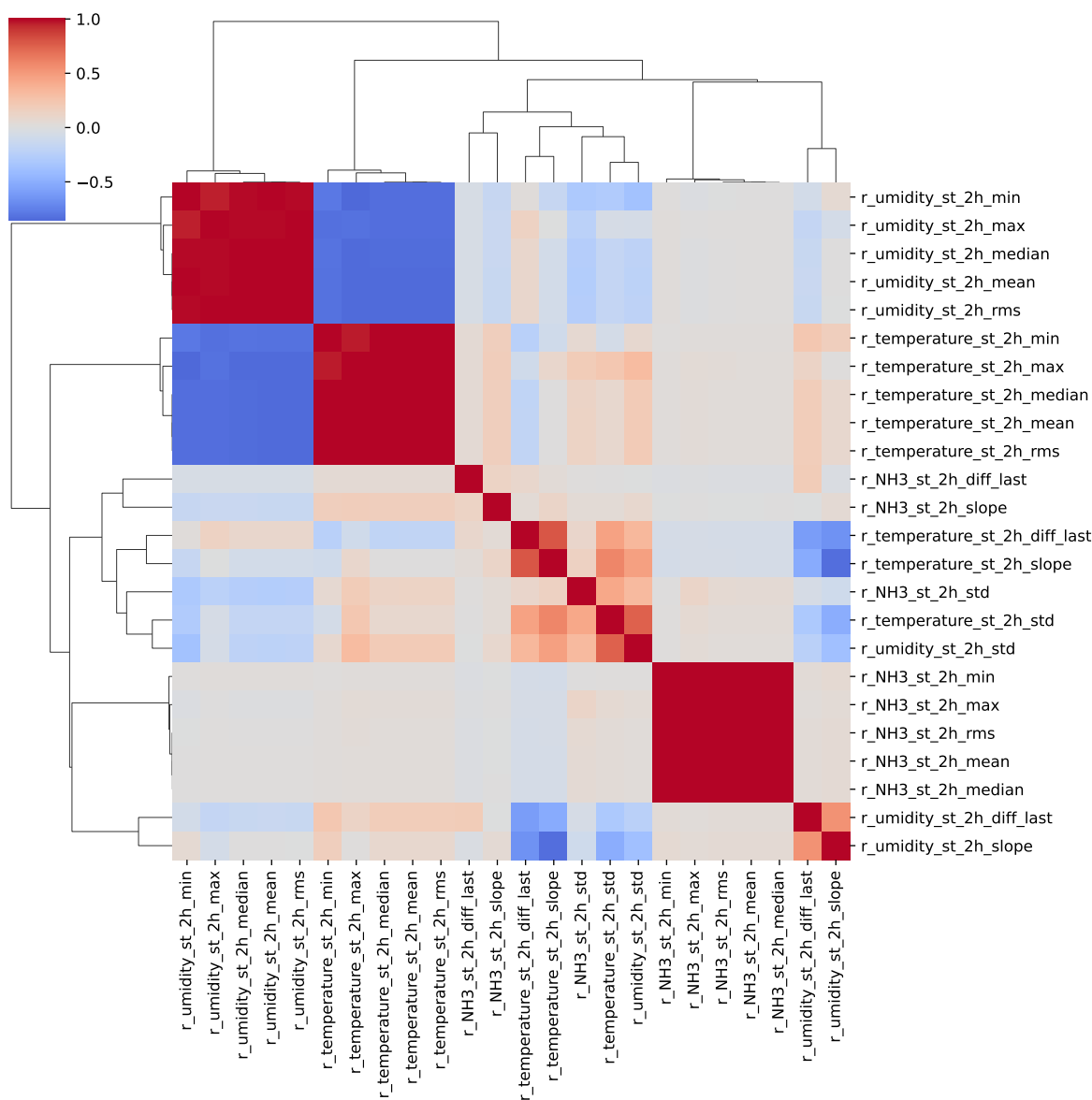
The following section has been done with a threshold choosen of : 0.8

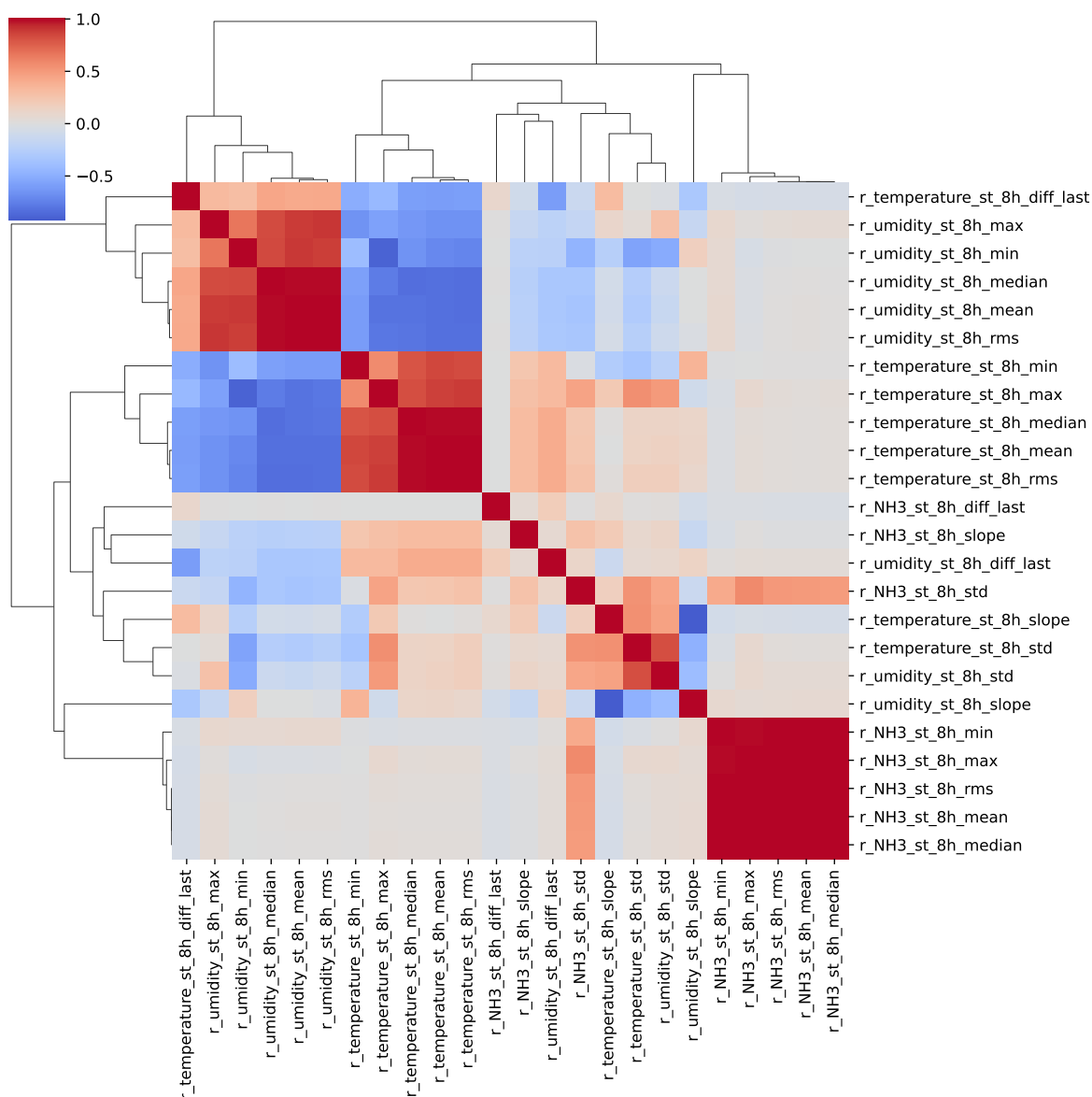
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_st_2h_min, r_NH3_st_2h_max, r_NH3_st_2h_median, r_NH3_st_2h_mean, r_NH3_st_2h_rms ; **cluster 7:** r_NH3_st_2h_std ; **cluster 8:** r_NH3_st_2h_diff_last ; **cluster 9:** r_NH3_st_2h_slope ; **cluster 10:** r_temperature_st_2h_min, r_umidity_st_2h_rms, r_temperature_st_2h_max, r_umidity_st_2h_max, r_temperature_st_2h_mean, r_umidity_st_2h_median, r_umidity_st_2h_mean, r_temperature_st_2h_rms, r_temperature_st_2h_median, r_umidity_st_2h_min ; **cluster 11:** r_temperature_st_2h_std ; **cluster 12:** r_temperature_st_2h_diff_last ; **cluster 13:** r_temperature_st_2h_slope, r_umidity_st_2h_slope ; **cluster 14:** r_umidity_st_2h_std ; **cluster 15:** r_umidity_st_2h_diff_last ; **cluster 16:** r_NH3_st_8h_min, r_NH3_st_8h_median, r_NH3_st_8h_max, r_NH3_st_8h_rms, r_NH3_st_8h_mean ; **cluster 17:** r_NH3_st_8h_std ; **cluster 18:** r_NH3_st_8h_diff_last ; **cluster 19:** r_NH3_st_8h_slope ; **cluster 20:** r_umidity_st_8h_min, r_umidity_st_8h_rms, r_umidity_st_8h_max, r_temperature_st_8h_rms, r_temperature_st_8h_min, r_temperature_st_8h_median, r_umidity_st_8h_mean, r_umidity_st_8h_median, r_temperature_st_8h_max, r_temperature_st_8h_mean ; **cluster 21:** r_temperature_st_8h_std, r_umidity_st_8h_std ; **cluster 22:** r_temperature_st_8h_diff_last ; **cluster 23:** r_umidity_st_8h_slope, r_temperature_st_8h_slope ; **cluster 24:** r_umidity_st_8h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features__X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_min, r_NH3_st_2h_std, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_temperature_st_2h_min, r_temperature_st_2h_std, r_temperature_st_2h_diff_last, r_temperature_st_2h_slope, r_umidity_st_2h_std, r_umidity_st_2h_diff_last, r_NH3_st_8h_min, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_umidity_st_8h_min, r_temperature_st_8h_std, r_temperature_st_8h_diff_last, r_umidity_st_8h_slope, r_umidity_st_8h_diff_last





3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_min, r_NH3_st_2h_std, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_temperature_st_2h_min, r_temperature_st_2h_std, r_temperature_st_2h_diff_last, r_tempera-

ture_st_2h_slope, r_umidity_st_2h_std, r_umidity_st_2h_diff_last, r_NH3_st_8h_min, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_umidity_st_8h_min, r_temperature_st_8h_std, r_temperature_st_8h_diff_last, r_umidity_st_8h_slope, r_umidity_st_8h_diff_last

Len(Features_X): 24

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

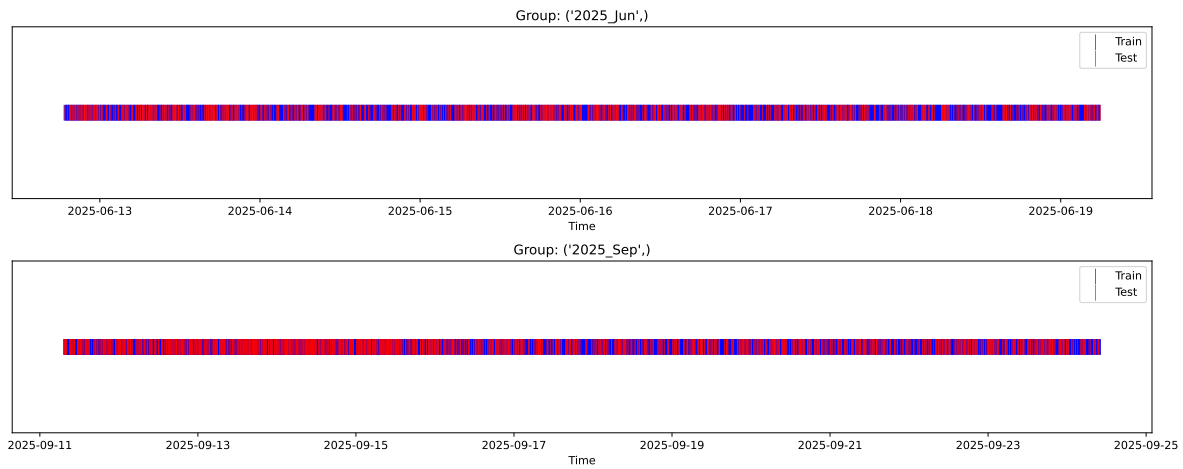
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 5513 samples Test: 2363 samples



3.3.2 Internal CV Folds

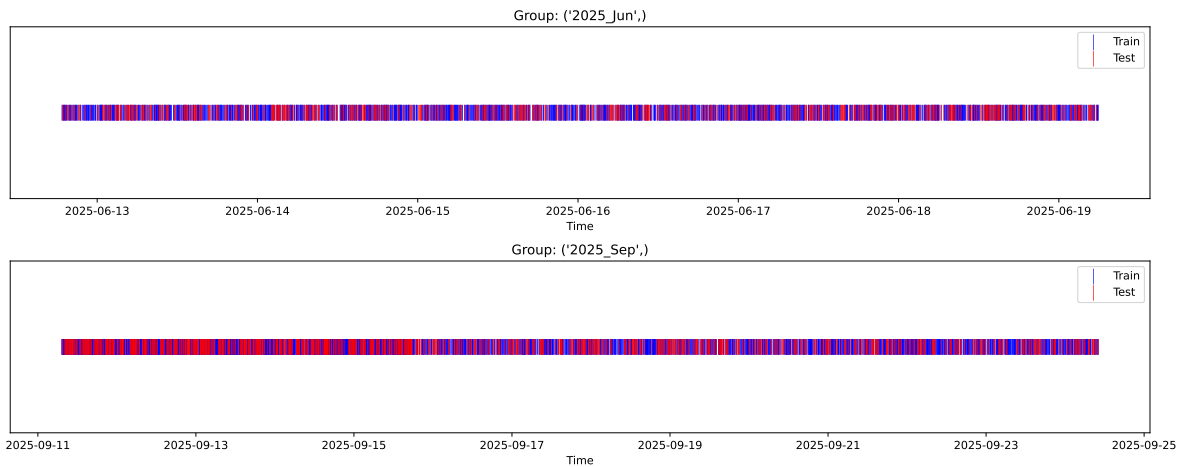
Fold 1/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



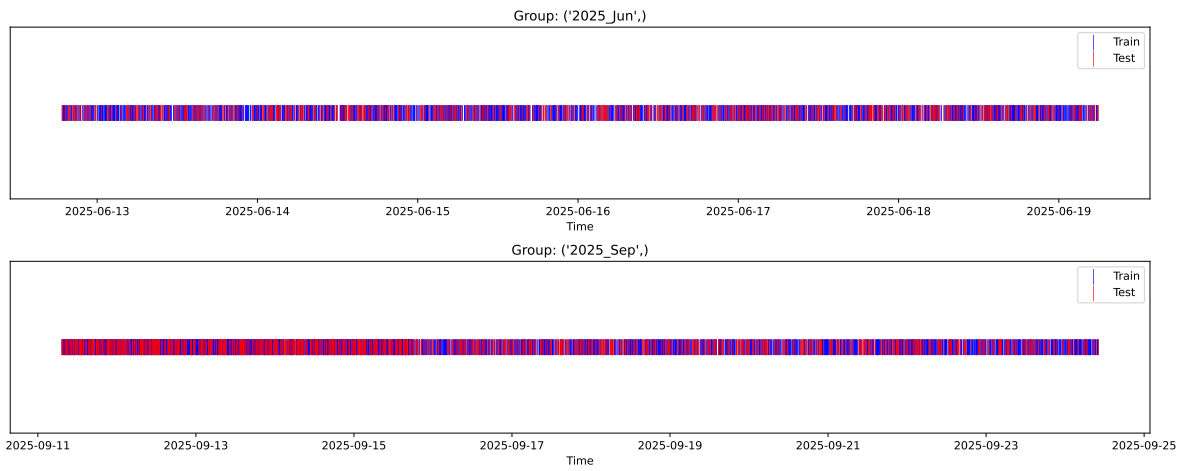
Fold 4/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3859 samples Test: 1654 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.354, std=0.811, min=0.959, max=13.845
 y_{test} : mean=2.358, std=0.777, min=1.068, max=7.822

Fold 0 — y_{test} : mean=2.35, std=0.80, y_{train} : mean=2.35, std=0.82

Fold 1 — y_{test} : mean=2.32, std=0.78, y_{train} : mean=2.37, std=0.82

Fold 2 — y_{test} : mean=2.35, std=0.85, y_{train} : mean=2.35, std=0.79

Fold 3 — y_test: mean=2.36, std=0.78, y_train: mean=2.35, std=0.83

Fold 4 — y_test: mean=2.32, std=0.75, y_train: mean=2.37, std=0.84

train scores CV: [0.54986911 0.55478546 0.57037035 0.55655692 0.55279794]

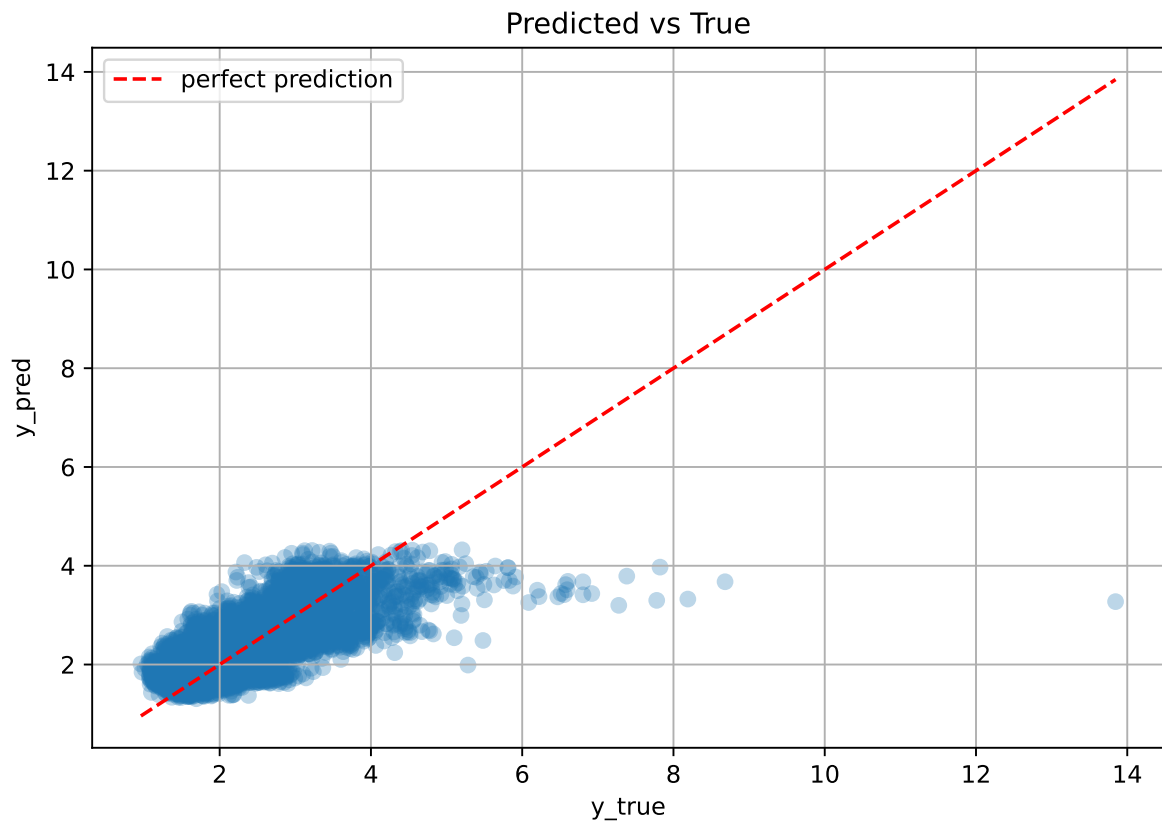
test scores CV: [0.57133455 0.55811323 0.52652331 0.55453444 0.56350634]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5569	0.5548	0.0152	0.5934	0.5951	0.0403	1.0726	1.0037

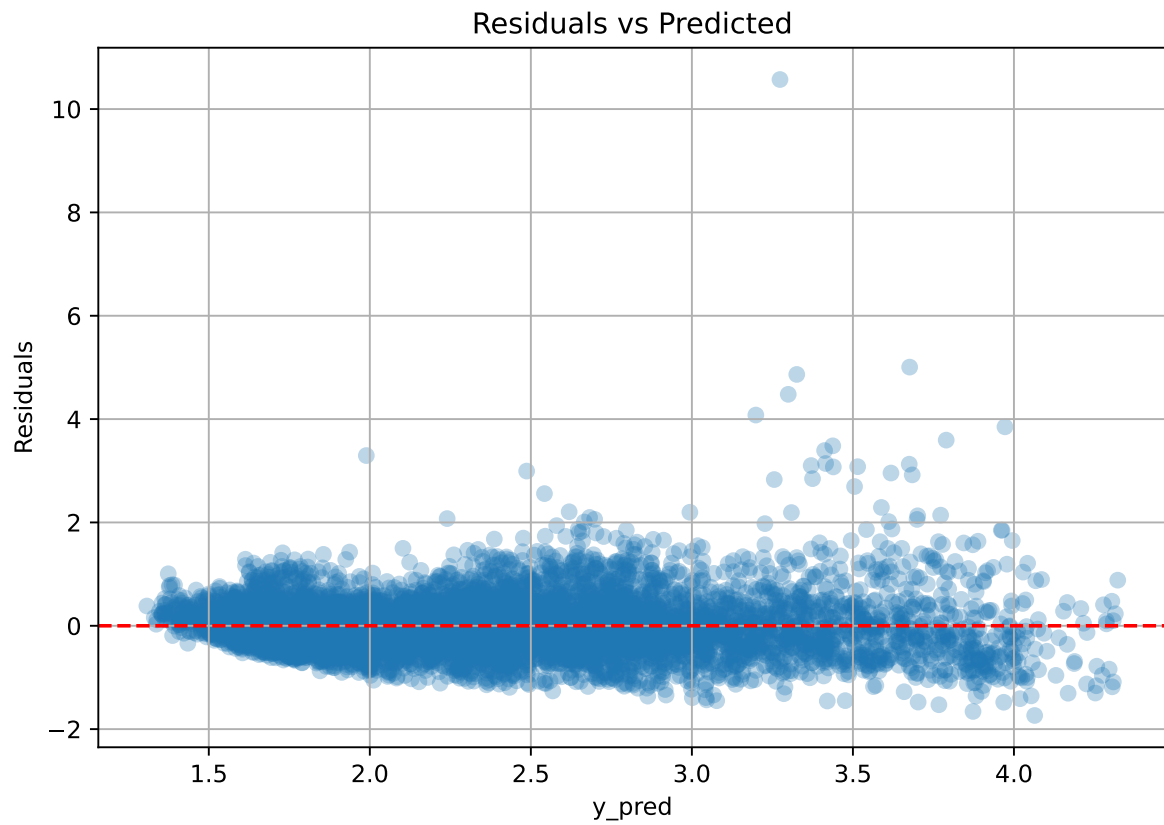
Coefficients:

coef_r_CO2 = -0.0475, coef_r_NH3 = -0.0120, coef_r_NH3_st_2h_diff_last = 0.0080, coef_r_NH3_st_2h_min = 0.2002, coef_r_NH3_st_2h_slope = -0.0126, coef_r_NH3_st_2h_std = 0.0497, coef_r_NH3_st_8h_diff_last = 0.0080, coef_r_NH3_st_8h_min = -0.2483, coef_r_NH3_st_8h_slope = 0.0040, coef_r_NH3_st_8h_std = 0.0111, coef_r_THI = 0.3571, coef_r_temperature = 0.9548, coef_r_temperature_st_2h_diff_last = -0.0179, coef_r_temperature_st_2h_min = -1.2487, coef_r_temperature_st_2h_slope = -0.1409, coef_r_temperature_st_2h_std = -0.1007, coef_r_temperature_st_8h_diff_last = -0.0179, coef_r_temperature_st_8h_std = 0.0571, coef_r_umidity = -0.6437, coef_r_umidity_st_2h_diff_last = -0.0043, coef_r_umidity_st_2h_std = 0.0612, coef_r_umidity_st_8h_diff_last = -0.0043, coef_r_umidity_st_8h_min = 0.2621, coef_r_umidity_st_8h_slope = 0.1984, intercept = 2.3540,

3.4.2 Scatter



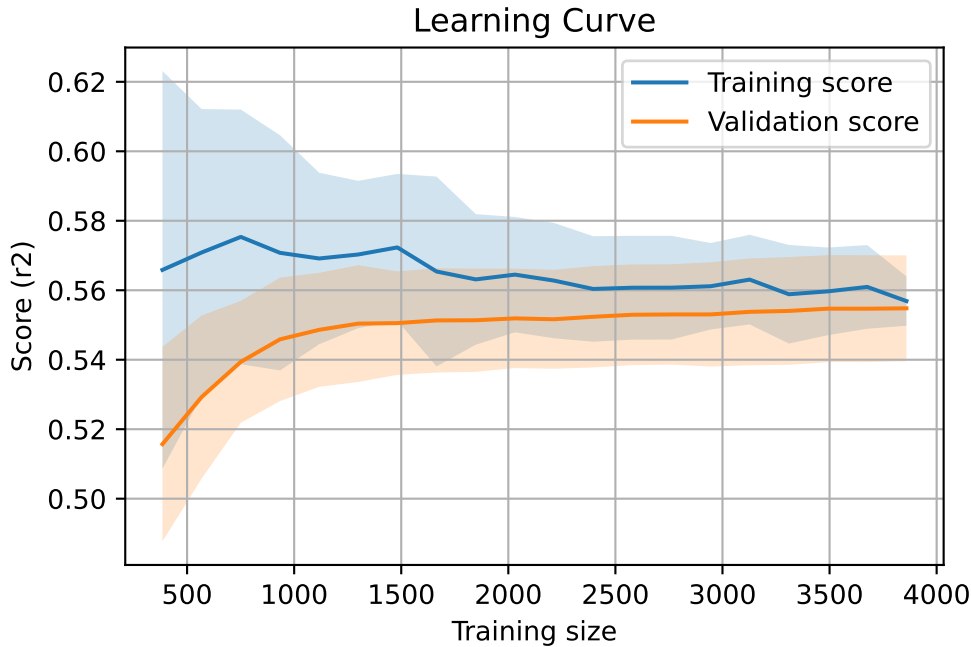
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7876, `Len(training)` : 5514,

`Len(training_real)` : 3860, `Len(test_of_training)` : 1654, `Len(test)` : 2362



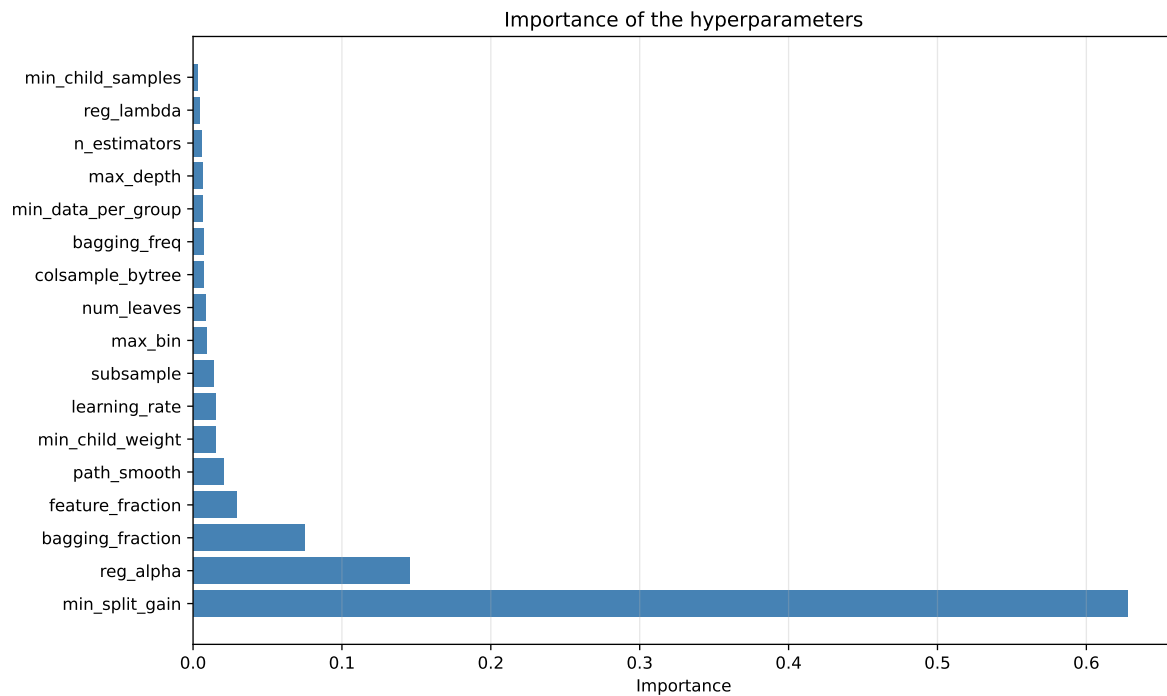
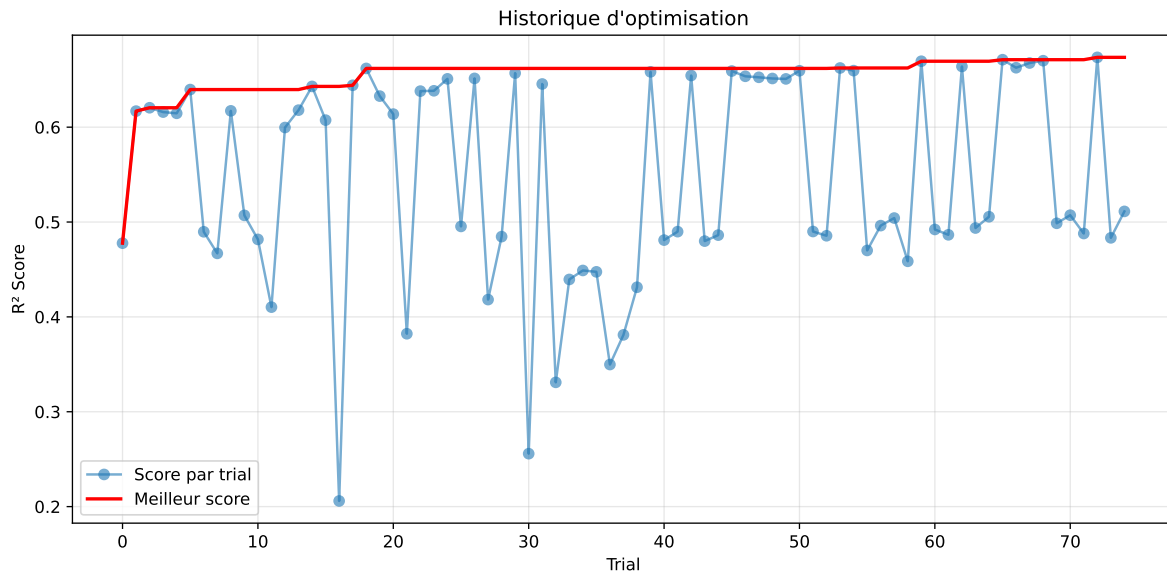
3.5 Model : LGBM

3.5.1 Gridsearch

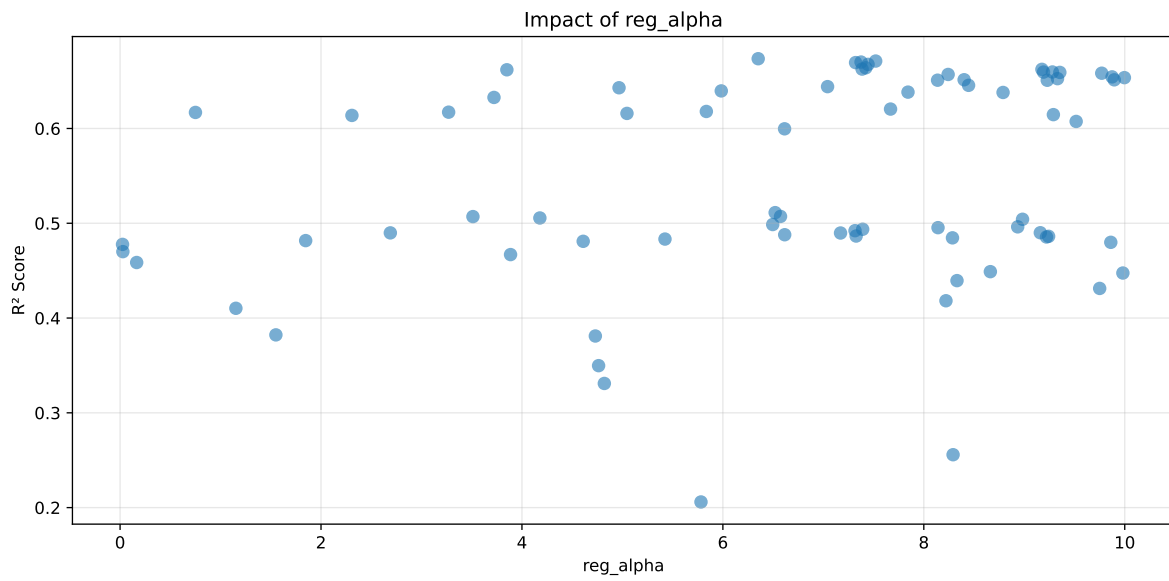
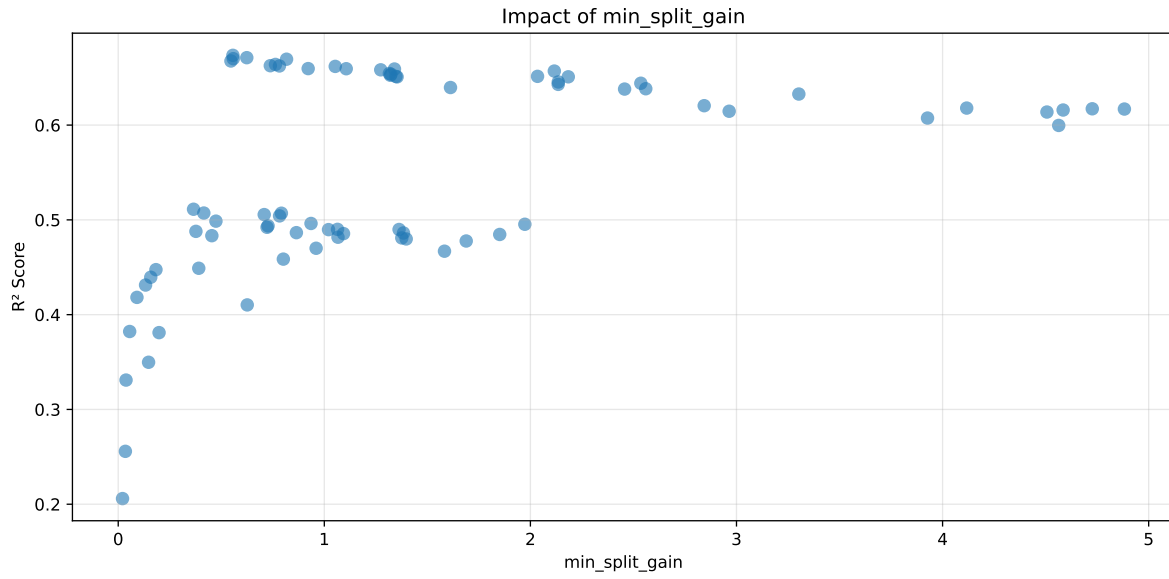
Distribution y_train vs y_test: y_train: mean=2.354, std=0.811, min=0.959, max=13.845
y_test: mean=2.358, std=0.777, min=1.068, max=7.822

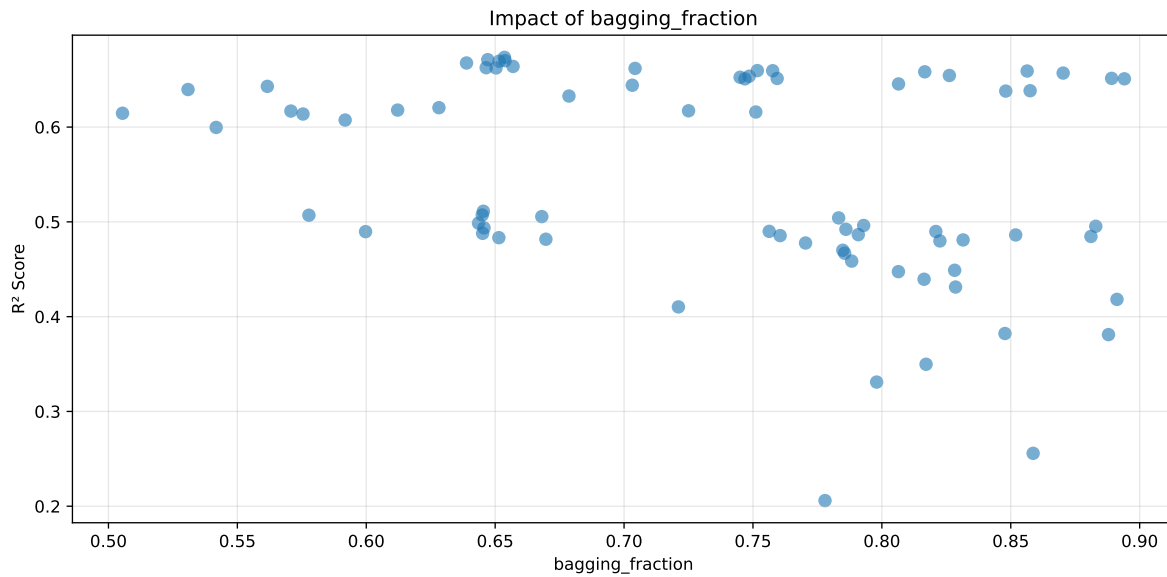
===== TOP Importance FEATURES =====

feature importance r_CO2 147 r_umidity 91 r_NH3_st_8h_min 89 r_NH3 74 r_temperature_st_2h_slope 63 r_umidity_st_8h_min 61 r_NH3_st_8h_std 56 r_NH3_st_2h_std 55 r_NH3_st_8h_slope 44 r_temperature_st_8h_std 43 r_umidity_st_8h_slope 37 r_temperature_st_2h_std 37 r_umidity_st_2h_std 36 r_THI 36 r_temperature_st_2h_diff_last 34 r_NH3_st_2h_min 31 r_NH3_st_2h_diff_last 31 r_temperature 31 r_umidity_st_2h_diff_last 23 r_NH3_st_2h_slope 22 r_temperature_st_2h_min 17 r_NH3_st_8h_diff_last 14 r_temperature_st_8h_diff_last 13 r_umidity_st_8h_diff_last 7



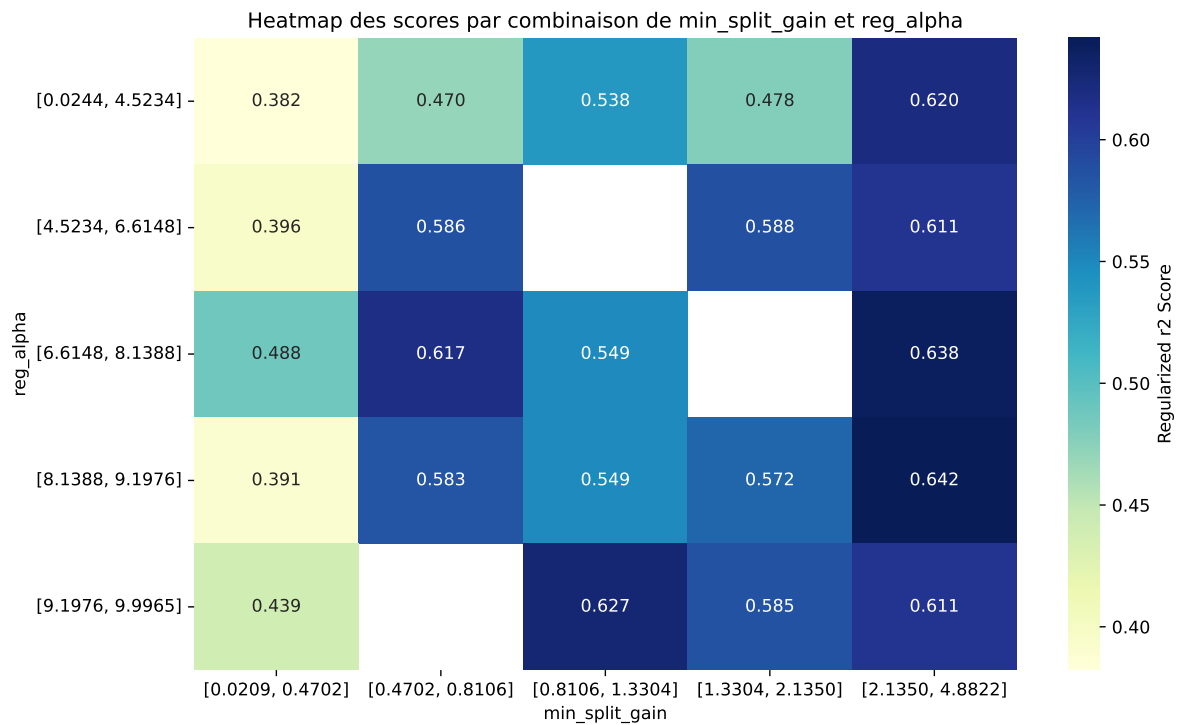
Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha', 'bagging_fraction']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7045	0.6736	0.7037	0.0189	0.6736	0.7172	0.0436	1.0647	1.0458
2	0.7038	0.6711	0.7021	0.0201	0.6711	0.7239	0.0528	1.0787	1.0487
3	0.7032	0.6701	0.7003	0.0152	0.6701	0.7196	0.0495	1.0739	1.0495
4	0.7007	0.6695	0.7002	0.0205	0.6695	0.7279	0.0584	1.0873	1.0466
5	0.6989	0.6676	0.6999	0.0173	0.6676	0.7216	0.0539	1.0808	1.0468
6	0.6927	0.664	0.6973	0.021	0.664	0.7136	0.0496	1.0747	1.0432
7	0.6904	0.6627	0.6946	0.02	0.6627	0.7128	0.0501	1.0756	1.0418
8	0.6929	0.6624	0.6932	0.0197	0.6624	0.7119	0.0496	1.0748	1.0461
9	0.6949	0.6619	0.693	0.0166	0.6619	0.716	0.054	1.0816	1.0498
10	0.6872	0.6596	0.6922	0.0184	0.6596	0.716	0.0564	1.0855	1.0418
---	---	---	---	---	---	---	---	---	---
75	0.8722	0.7612	0.78	0.0235	0.2059	0.7995	0.0383	1.0503	1.1459

Hyperparameters:

Rank 1: {'reg__n_estimators': 400, 'reg__max_depth': 3, 'reg__learning_rate': 0.05593826194028376, 'reg__num_leaves': 60, 'reg__subsample': 0.8281576148604917, 'reg__colsample_bytree': 0.7446555847288359, 'reg__min_child_samples': 14, 'reg__reg_alpha': 6.351563646228377, 'reg__reg_lambda': 46.797479971318, 'reg__min_split_gain': 0.5564154091968744, 'reg__path_smooth': 2.7659622146297624, 'reg__min_child_weight': 0.006094682786641304, 'reg__feature_fraction': 0.5427356691340816, 'reg__bagging_fraction': 0.6536110811950746, 'reg__bagging_freq': 5, 'reg__max_bin': 78, 'reg__min_data_per_group': 134}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.07347948113218014, 'reg__num_leaves': 60, 'reg__subsample': 0.7864246909364495, 'reg__colsample_bytree': 0.685497643409828, 'reg__min_child_samples': 16, 'reg__reg_alpha': 7.5209694797745374, 'reg__reg_lambda': 34.470561203095016, 'reg__min_split_gain': 0.6238338022743659, 'reg__path_smooth': 2.824130099930527, 'reg__min_child_weight': 0.014611083922813663, 'reg__feature_fraction': 0.5737880718893643, 'reg__bagging_fraction': 0.6471224256362496, 'reg__bagging_freq': 5, 'reg__max_bin': 82, 'reg__min_data_per_group': 130}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.07421464493446676, 'reg__num_leaves': 60, 'reg__subsample': 0.7794828488543621, 'reg__colsample_bytree': 0.6767032403262885, 'reg__min_child_samples': 16, 'reg__reg_alpha': 7.375302084057597, 'reg__reg_lambda': 32.062463581735756, 'reg__min_split_gain': 0.5581549841550293, 'reg__path_smooth': 2.8295640919083986, 'reg__min_child_weight': 2.429789325463443, 'reg__feature_fraction': 0.5434732443472057, 'reg__bagging_fraction': 0.653844794471909, 'reg__bagging_freq': 5, 'reg__max_bin': 81, 'reg__min_data_per_group': 130}

Rank 4: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.0847204453766534, 'reg__num_leaves': 29, 'reg__subsample': 0.7955073488721986, 'reg__colsample_bytree': 0.7132305388494169, 'reg__min_child_samples': 19, 'reg__reg_alpha': 7.320779967873391, 'reg__reg_lambda': 34.517741180801664, 'reg__min_split_gain': 0.8168299530833708, 'reg__path_smooth': 2.1983665206074186, 'reg__min_child_weight': 2.020472563989255, 'reg__feature_fraction': 0.5829153751393519, 'reg__bagging_fraction': 0.6515058513086694, 'reg__bagging_freq': 4, 'reg__max_bin': 63, 'reg__min_data_per_group': 152}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.07325476497087913, 'reg__num_leaves': 56, 'reg__subsample': 0.7757543139604638, 'reg__colsample_bytree': 0.7452549618636274, 'reg__min_child_samples': 15, 'reg__reg_alpha': 7.445081572960934, 'reg__reg_lambda': 45.72833317235607, 'reg__min_split_gain': 0.5469201523818787, 'reg__path_smooth': 2.8931783920824716, 'reg__min_child_weight': 2.804110216113814, 'reg__feature_fraction': 0.579934308608671, 'reg__bagging_fraction': 0.6388837601951356, 'reg__bagging_freq': 5, 'reg__max_bin': 85, 'reg__min_data_per_group': 128}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.08486109731005652, 'reg__num_leaves': 30, 'reg__subsample': 0.7828311536696348, 'reg__colsample_bytree': 0.6807074720895113, 'reg__min_child_samples': 32, 'reg__reg_alpha': 7.422444848393919, 'reg__reg_lambda': 46.78402822524895, 'reg__min_split_gain': 0.7630078046522843, 'reg__path_smooth': 2.7572761801631067, 'reg__min_child_weight': 0.011126319460694704, 'reg__feature_fraction': 0.5893446565622555, 'reg__bagging_fraction': 0.6569401654640445, 'reg__bagging_freq': 4, 'reg__max_bin': 65, 'reg__min_data_per_group': 131}

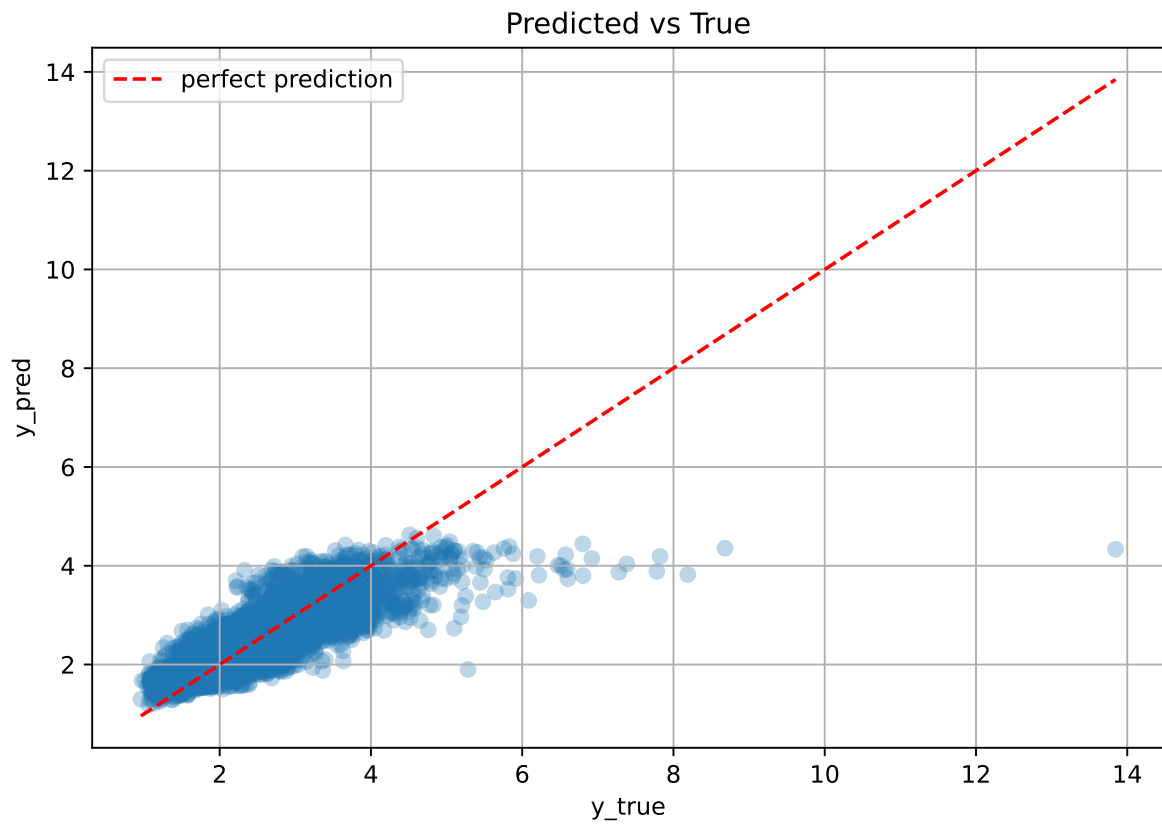
Rank 7: {'reg__n_estimators': 400, 'reg__max_depth': 3, 'reg__learning_rate': 0.07525433483735523, 'reg__num_leaves': 60, 'reg__subsample': 0.7798789785067654, 'reg__colsample_bytree': 0.7472395842985975, 'reg__min_child_samples': 32, 'reg__reg_alpha': 7.38581277067679, 'reg__reg_lambda': 46.59824769983587, 'reg__min_split_gain': 0.7379543481000826, 'reg__path_smooth': 2.750691502709328, 'reg__min_child_weight': 0.047392840185735995, 'reg__feature_fraction': 0.5915250857301964, 'reg__bagging_fraction': 0.6465138237258135, 'reg__bagging_freq': 4, 'reg__max_bin': 87, 'reg__min_data_per_group': 131}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.07140225081522998, 'reg__num_leaves': 29, 'reg__subsample': 0.8056356703372868, 'reg__colsample_bytree': 0.7294079837124328, 'reg__min_child_samples': 20, 'reg__reg_alpha': 9.175608588292846, 'reg__reg_lambda': 42.89394918580217, 'reg__min_split_gain': 0.7821480172239956, 'reg__path_smooth': 2.2040870137802235, 'reg__min_child_weight': 0.11319720126316637, 'reg__feature_fraction': 0.5284443247549786, 'reg__bagging_fraction': 0.6503520021482379, 'reg__bagging_freq': 4, 'reg__max_bin': 125, 'reg__min_data_per_group': 156}

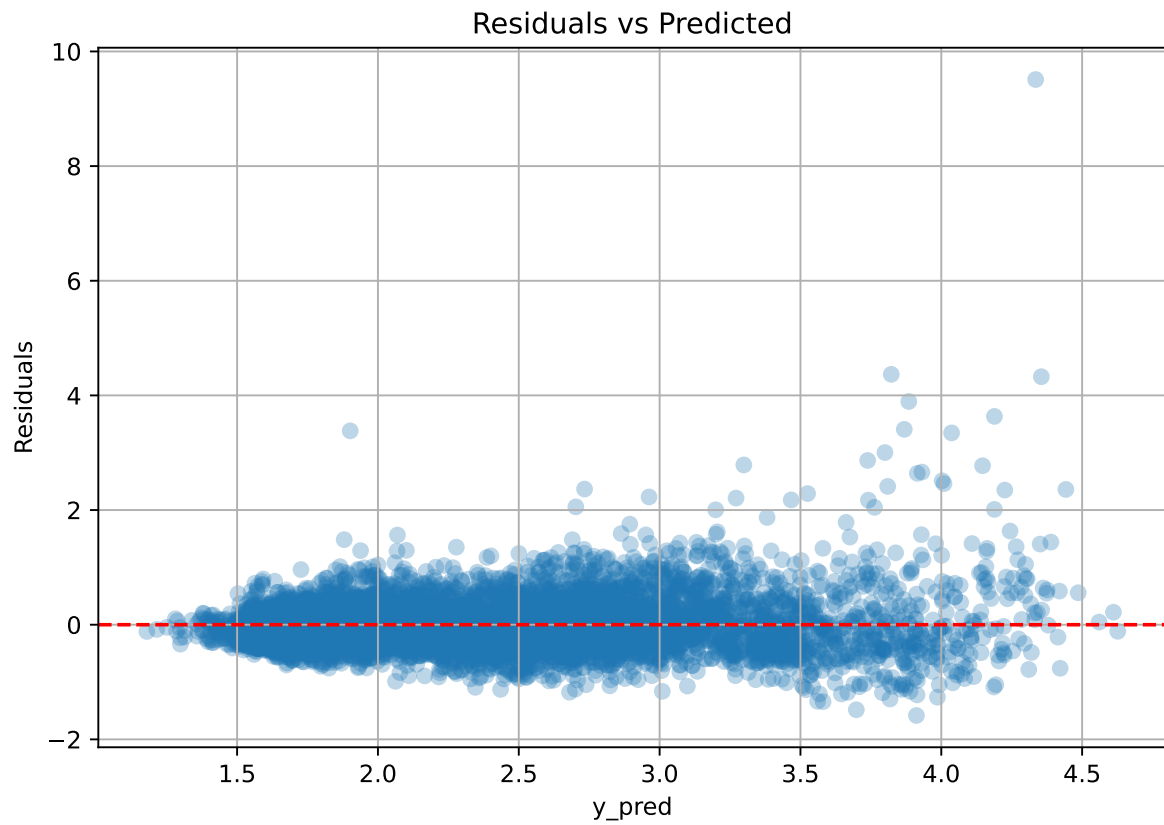
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.07409814625818421, 'reg__num_leaves': 58, 'reg__subsample': 0.874619950610696, 'reg__colsample_bytree': 0.7654453061011581, 'reg__min_child_samples': 16, 'reg__reg_alpha': 3.8499473758288727, 'reg__reg_lambda': 45.63991176559817, 'reg__min_split_gain': 1.0530200335136968, 'reg__path_smooth': 2.362776361867494, 'reg__min_child_weight': 0.006144013505899227, 'reg__feature_fraction': 0.888143388591023, 'reg__bagging_fraction': 0.7042733785898072, 'reg__bagging_freq': 3, 'reg__max_bin': 63, 'reg__min_data_per_group': 183}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.07098750716993113, 'reg__num_leaves': 63, 'reg__subsample': 0.8100047563527906, 'reg__colsample_bytree': 0.7178033542965999, 'reg__min_child_samples': 32, 'reg__reg_alpha': 9.280170036050666, 'reg__reg_lambda': 43.13124488999951, 'reg__min_split_gain': 0.9216955729276071, 'reg__path_smooth': 0.911751438833506, 'reg__min_child_weight': 0.10828526289871042, 'reg__feature_fraction': 0.5282949362800392, 'reg__bagging_fraction': 0.7517013509604207, 'reg__bagging_freq': 4, 'reg__max_bin': 68, 'reg__min_data_per_group': 154}

3.5.2 Scatter



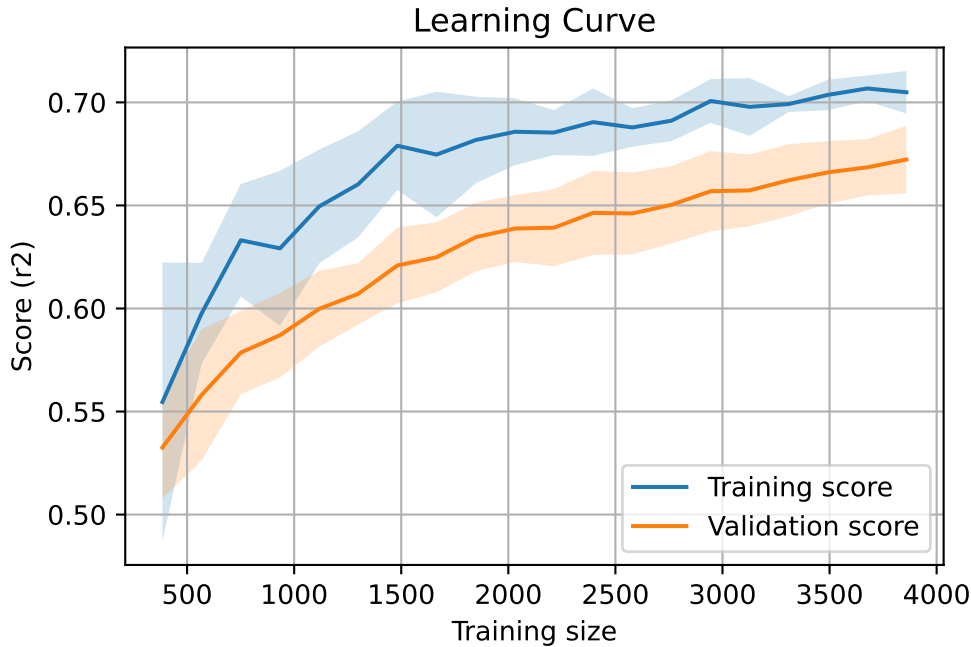
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7876, `Len(training)` : 5514,

`Len(training_real)` : 3860, `Len(test_of_training)` : 1654, `Len(test)` : 2362



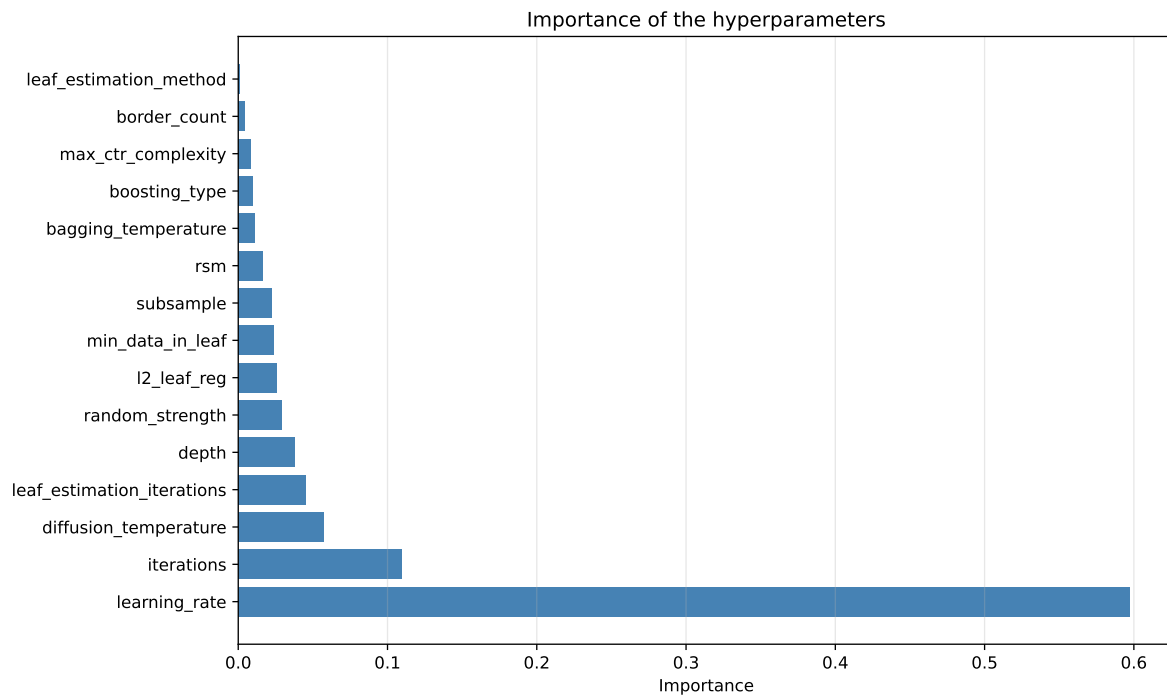
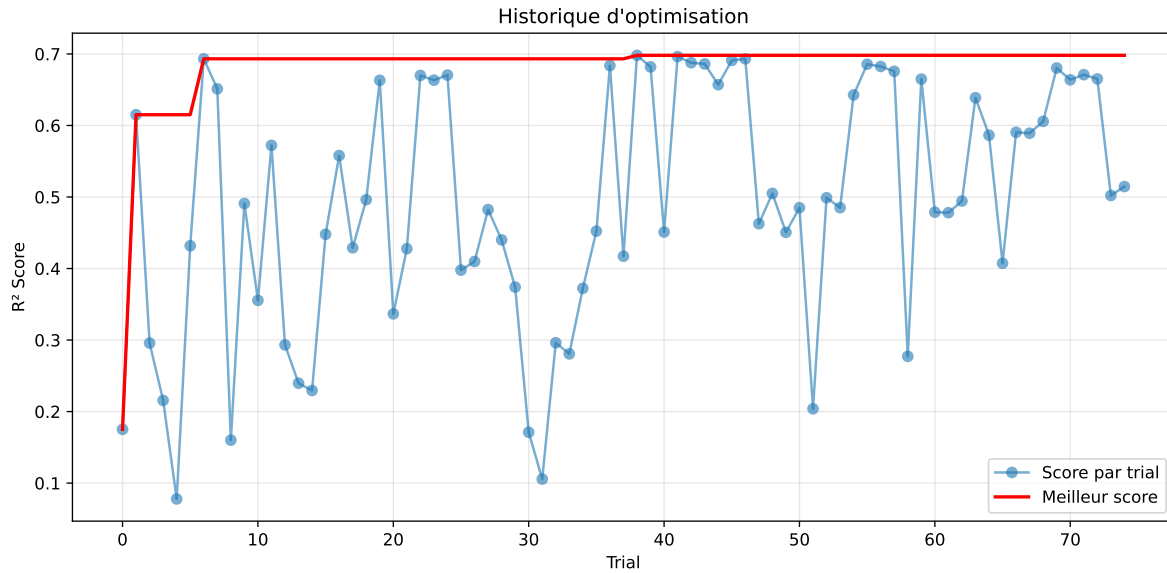
3.6 Model : CatBoost

3.6.1 Gridsearch

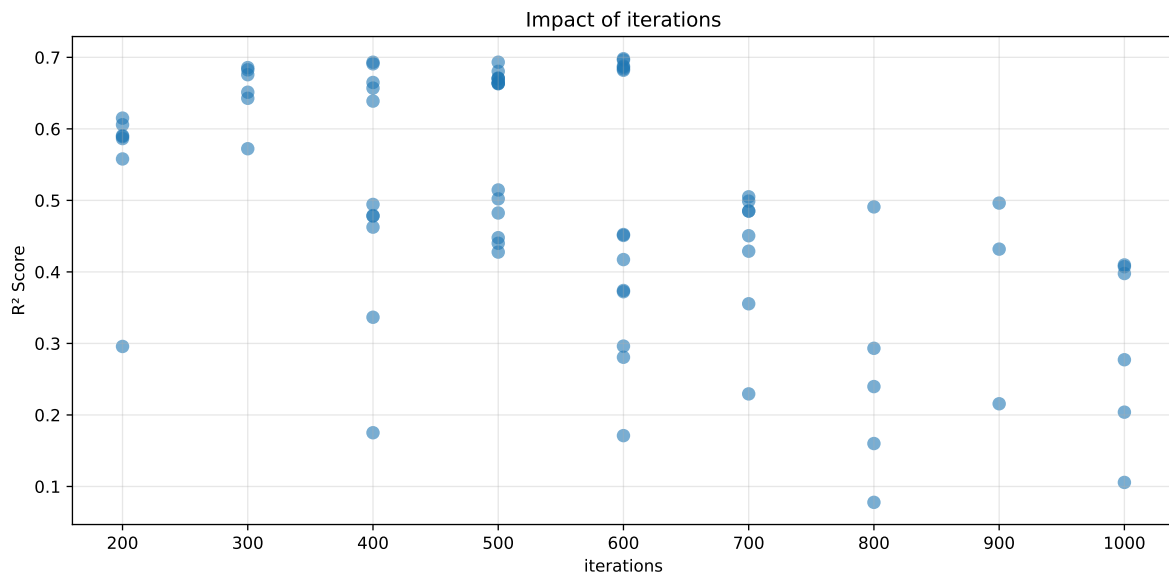
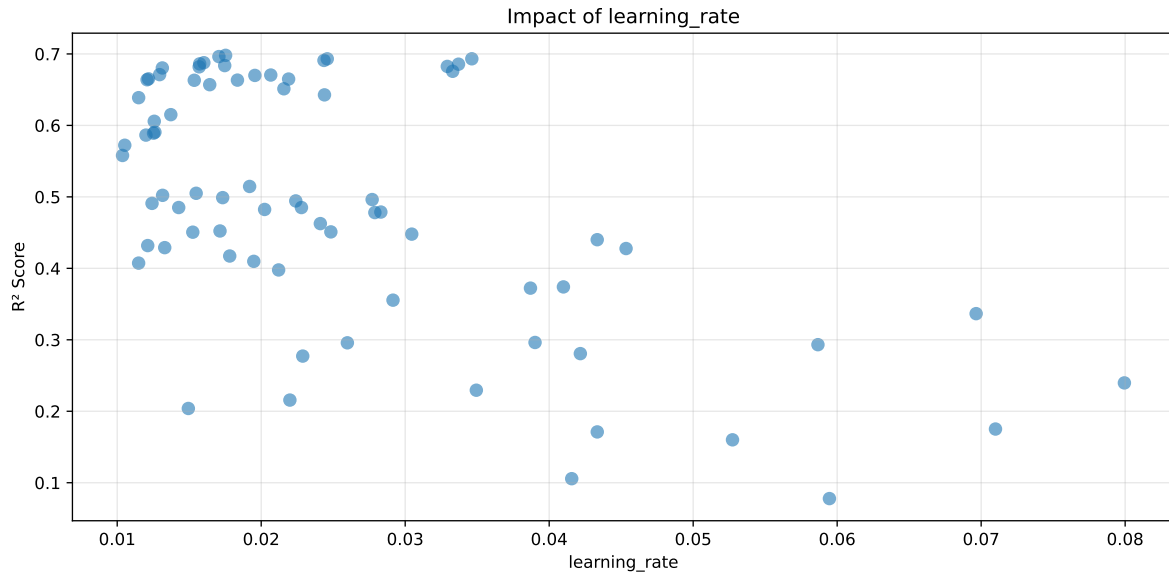
Distribution y_train vs y_test: y_train: mean=2.354, std=0.811, min=0.959, max=13.845
y_test: mean=2.358, std=0.777, min=1.068, max=7.822

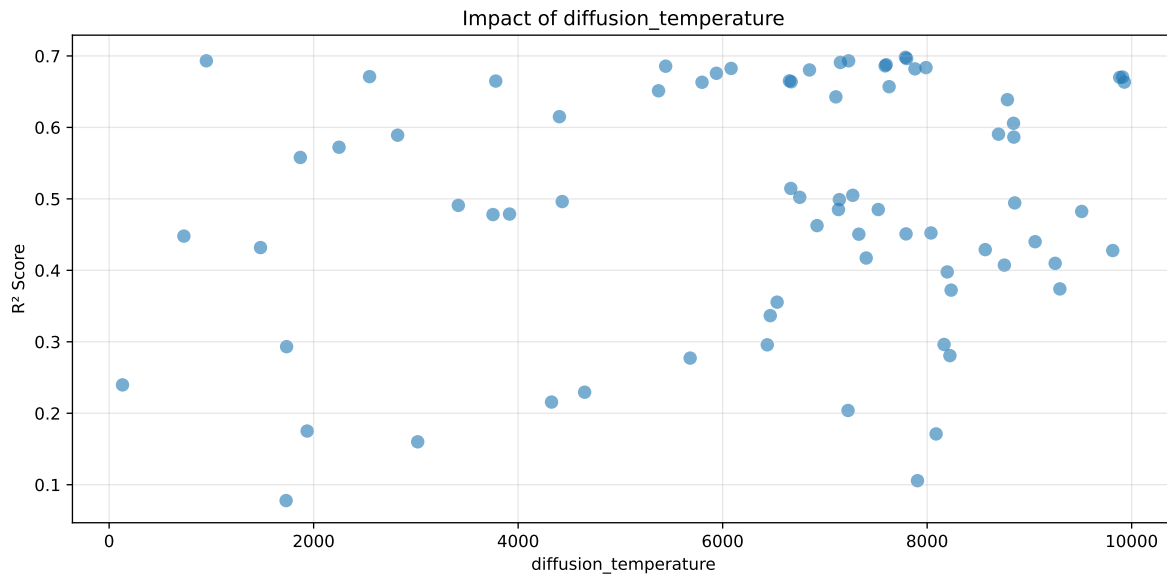
===== TOP Importance FEATURES =====

feature importance r_umidity 17.924406 r_temperature_st_2h_slope 11.437654 r_temperature_st_2h_std 9.754055 r_CO2 8.209170 r_umidity_st_8h_min 4.504493 r_temperature_st_8h_diff_last 4.061094 r_temperature_st_2h_diff_last 4.021672 r_NH3_st_8h_min 4.015966 r_temperature 3.877337 r_NH3 3.679446 r_umidity_st_8h_slope 3.624819 r_NH3_st_2h_std 3.085563 r_NH3_st_8h_std 3.001380 r_NH3_st_2h_slope 2.510431 r_NH3_st_2h_min 2.425411 r_NH3_st_8h_slope 2.410183 r_THI 2.268707 r_temperature_st_2h_min 2.157566 r_umidity_st_2h_std 1.848895 r_temperature_st_8h_std 1.580557 r_umidity_st_8h_diff_last 1.117371 r_NH3_st_2h_diff_last 0.872176 r_umidity_st_2h_diff_last 0.824594 r_NH3_st_8h_diff_last 0.787055



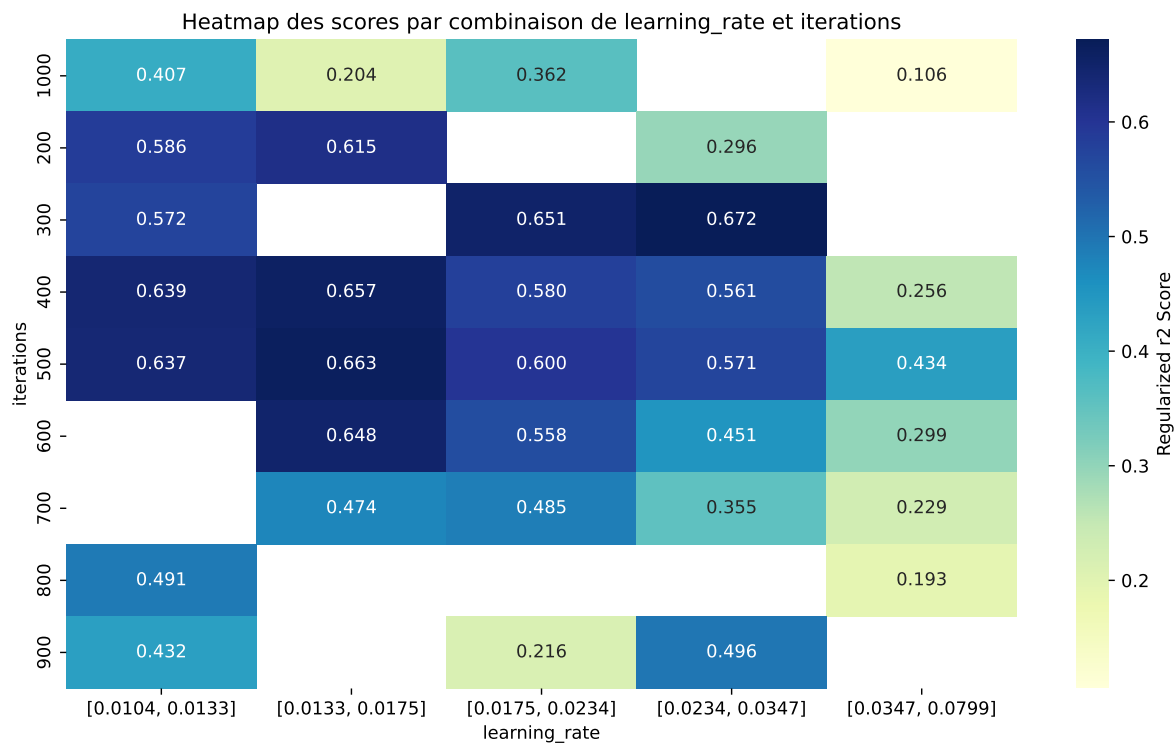
Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'diffusion_temperature']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7312	0.6981	0.7206	0.0189	0.6981	0.7261	0.028	1.0402	1.0474
2	0.7297	0.6963	0.7202	0.0207	0.6963	0.7281	0.0318	1.0457	1.048
3	0.7264	0.6932	0.7167	0.0176	0.6932	0.7333	0.0401	1.0578	1.0479
4	0.7263	0.6932	0.7165	0.02	0.6932	0.7225	0.0294	1.0423	1.0477
5	0.7229	0.6909	0.7153	0.0207	0.6909	0.7235	0.0325	1.0471	1.0462
6	0.7208	0.6877	0.7134	0.0193	0.6877	0.7203	0.0326	1.0474	1.048
7	0.7179	0.6861	0.7115	0.0193	0.6861	0.7193	0.0332	1.0484	1.0463
8	0.7152	0.6857	0.7107	0.0189	0.6857	0.7159	0.0303	1.0441	1.043
9	0.7111	0.6837	0.7088	0.0189	0.6837	0.7173	0.0335	1.049	1.04
10	0.7077	0.6825	0.7079	0.0197	0.6825	0.711	0.0285	1.0417	1.0369
---	---	---	---	---	---	---	---	---	---
75	0.9422	0.7982	0.8012	0.0231	0.0778	0.819	0.0208	1.0261	1.1805

Hyperparameters:

Rank 1: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.017527746210273275, 'reg__l2_leaf_reg': 23.47124636794813, 'reg__bagging_temperature': 0.5289311325687029, 'reg__random_strength': 4.26482169997875, 'reg__subsample': 0.601038206780298, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.8995120065488219, 'reg__border_count': 72, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7787.847530871148}

Rank 2: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.01706189059242447, 'reg__l2_leaf_reg': 24.288598878374383, 'reg__bagging_temperature': 0.5214497211784609, 'reg__random_strength': 3.3279015090699433, 'reg__subsample': 0.8312172609869041, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.6624792055297087, 'reg__border_count': 71, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7799.702006788757}

Rank 3: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.0346270736612356, 'reg__l2_leaf_reg': 40.49195783626053, 'reg__bagging_temperature': 4.027118714503331, 'reg__random_strength': 3.660988442147916, 'reg__subsample': 0.6006665391163282, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8008208571266608, 'reg__border_count': 115, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 950.8818241455552}

Rank 4: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.02458900716269521, 'reg__l2_leaf_reg': 21.627115058913855, 'reg__bagging_temperature': 4.225094752887897, 'reg__random_strength': 3.0897668366947078, 'reg__subsample': 0.8995704589535234, 'reg__min_data_in_leaf': 46, 'reg__leaf_estimation_iterations': 3, 'reg__rsm':

0.6627455730443553, 'reg__border_count': 92, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7231.923183019074}

Rank 5: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.024359235694869627, 'reg__l2_leaf_reg': 23.616935334182298, 'reg__bagging_temperature': 1.3117218423419175, 'reg__random_strength': 3.188267848274693, 'reg__subsample': 0.8827117429365388, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.6672591303818513, 'reg__border_count': 92, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7151.9097728655115}

Rank 6: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.016015720274071325, 'reg__l2_leaf_reg': 22.099818539687554, 'reg__bagging_temperature': 0.609732190922404, 'reg__random_strength': 3.3457359377658378, 'reg__subsample': 0.826969662262423, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.6658434248769675, 'reg__border_count': 69, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7599.71582709425}

Rank 7: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.015726562200039904, 'reg__l2_leaf_reg': 22.964280337168038, 'reg__bagging_temperature': 0.8220130550606153, 'reg__random_strength': 3.3359043263198584, 'reg__subsample': 0.8832841798843764, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.6619583323392639, 'reg__border_count': 70, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7589.082136830378}

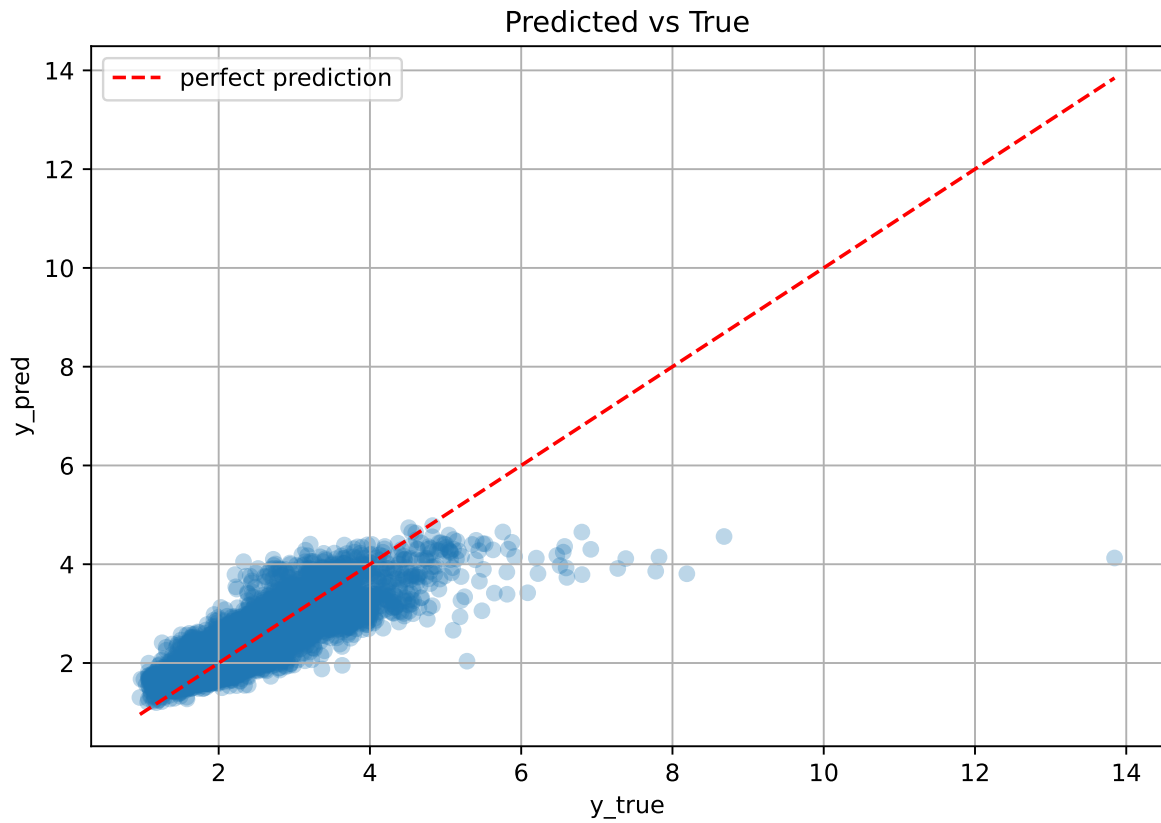
Rank 8: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.03370292439325006, 'reg__l2_leaf_reg': 18.6480909736376, 'reg__bagging_temperature': 1.757650313467438, 'reg__random_strength': 2.273943237533918, 'reg__subsample': 0.8205902284177955, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.6189242489289835, 'reg__border_count': 98, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5442.756699934345}

Rank 9: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.01746566513389897, 'reg__l2_leaf_reg': 41.77641249426253, 'reg__bagging_temperature': 0.5485210590030019, 'reg__random_strength': 4.296518481324035, 'reg__subsample': 0.7563314675939128, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.8085496348450835, 'reg__border_count': 72, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7988.750557507057}

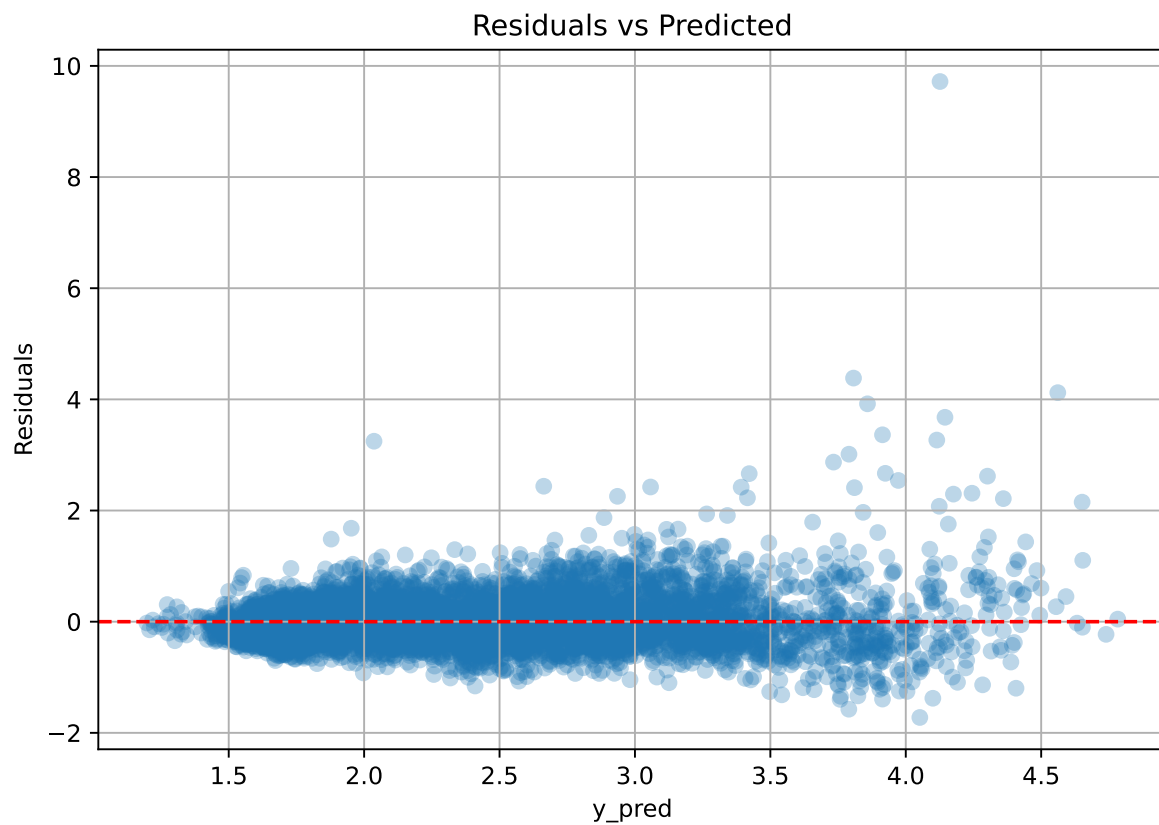
Rank 10: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.03292594393132751, 'reg__l2_leaf_reg': 17.913652414096333, 'reg__bagging_temperature': 4.971389501338977,

```
'reg__random_strength': 2.414522218278173, 'reg__subsample': 0.8419131029794362,  
'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 4, 'reg__rsm':  
0.620990998663248, 'reg__border_count': 41, 'reg__leaf_estimation_method': 'Newton',  
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_tempera-  
ture': 6082.770194020424}
```

3.6.2 Scatter



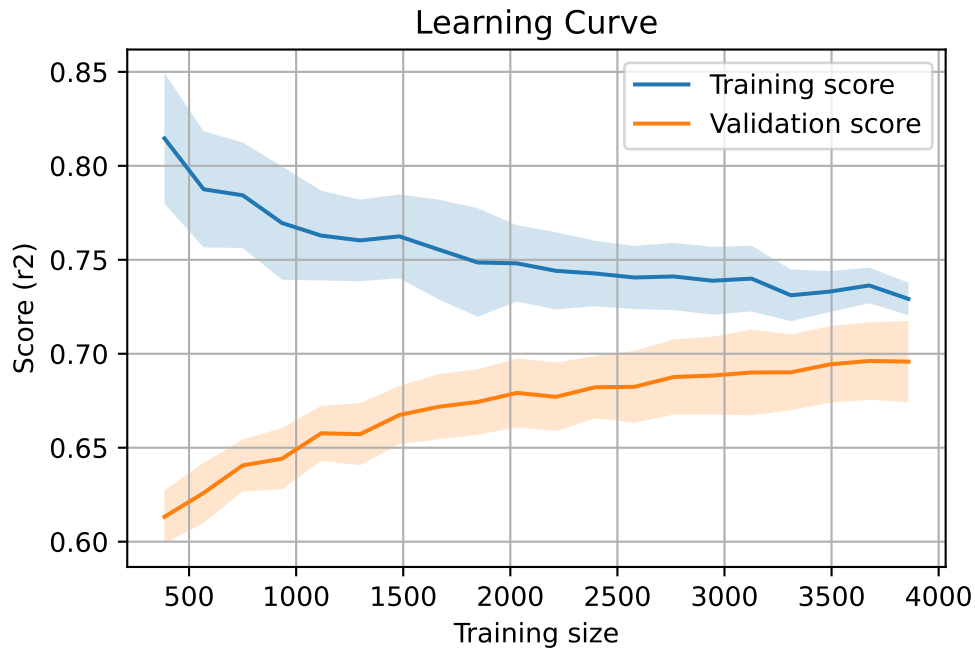
3.6.3 Scatter residuals



3.6.4 Learning curve

Len(group) : 7876, **Len(training)** : 5514,

Len(training_real) : 3860, **Len(test_of_training)** : 1654, **Len(test)** : 2362



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.557	0.555	0.015	0.593	0.595	0.04	1.073	1.004	nan
LGBM	0.704	0.674	0.019	0.704	0.717	0.044	1.065	1.046	0.6736
CatBoost	0.731	0.698	0.019	0.721	0.726	0.028	1.04	1.047	0.6981

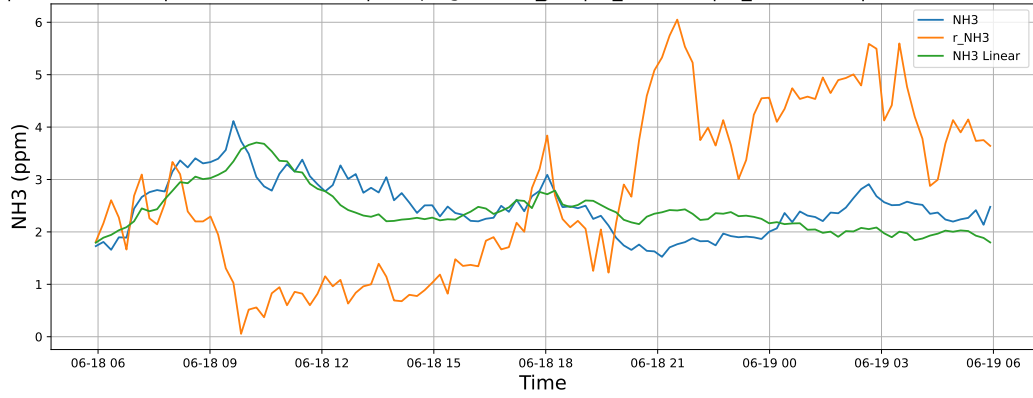
4 Plot

4.1 Standalone plots

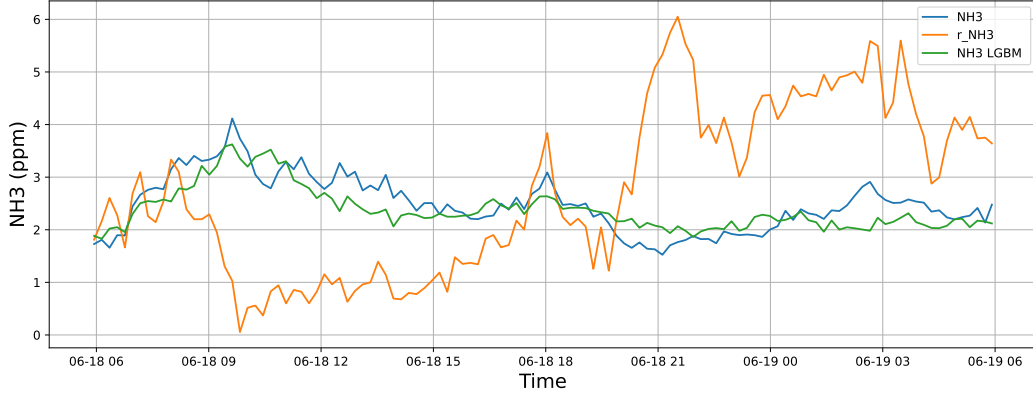
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

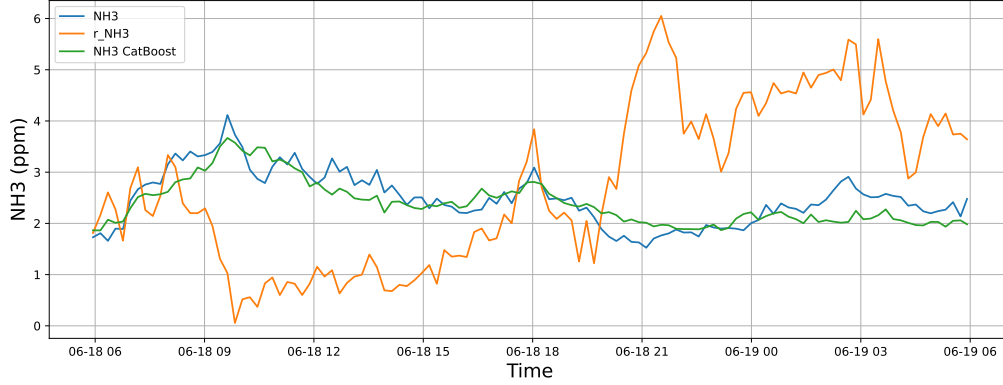
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

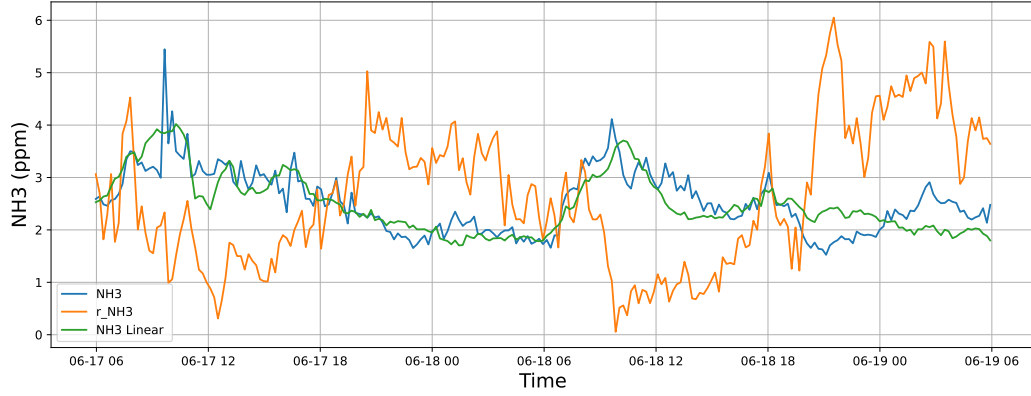


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

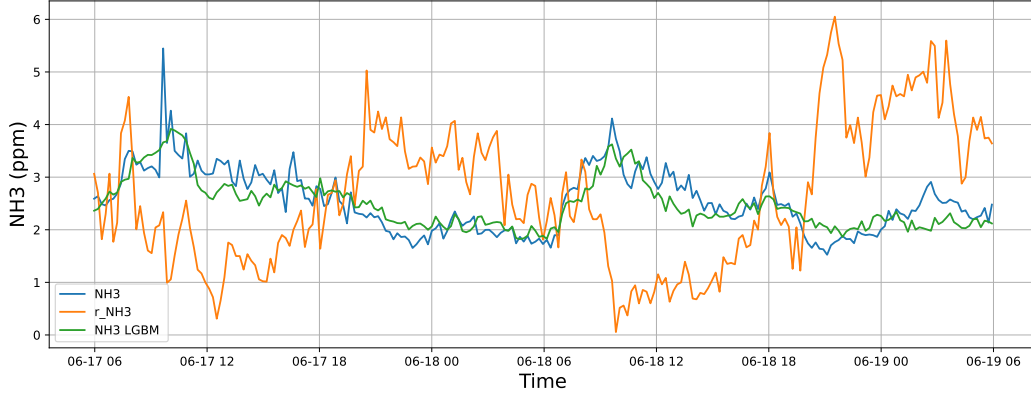


4.1.1.2 Time window : 48

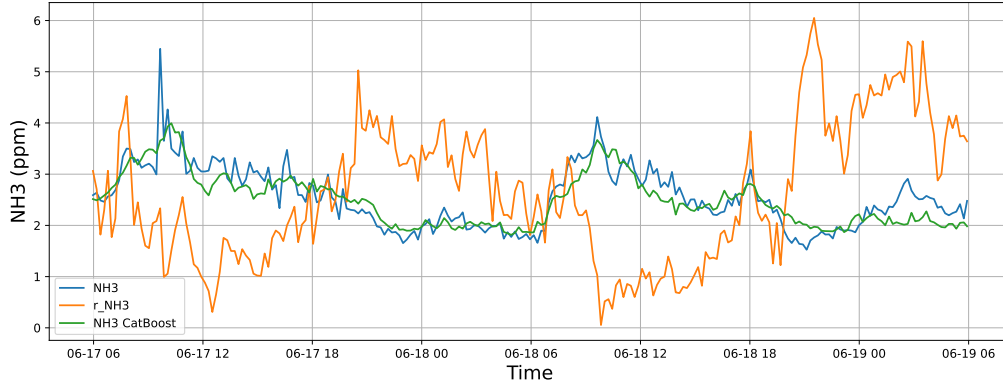
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

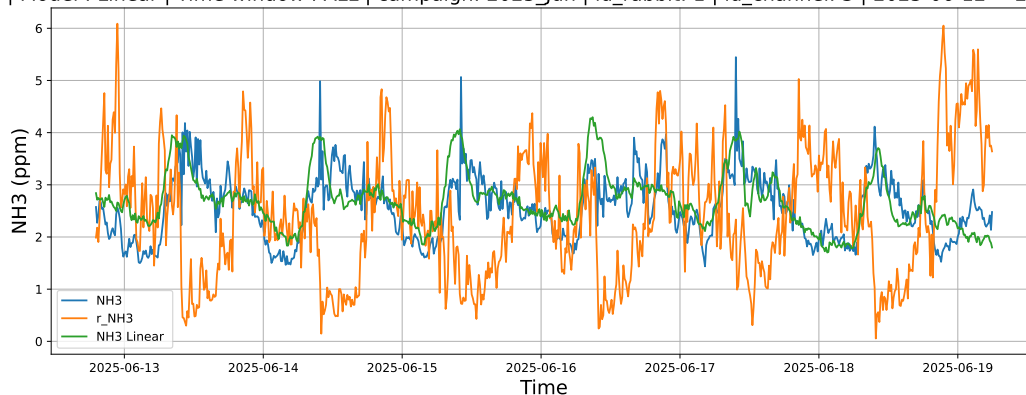


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

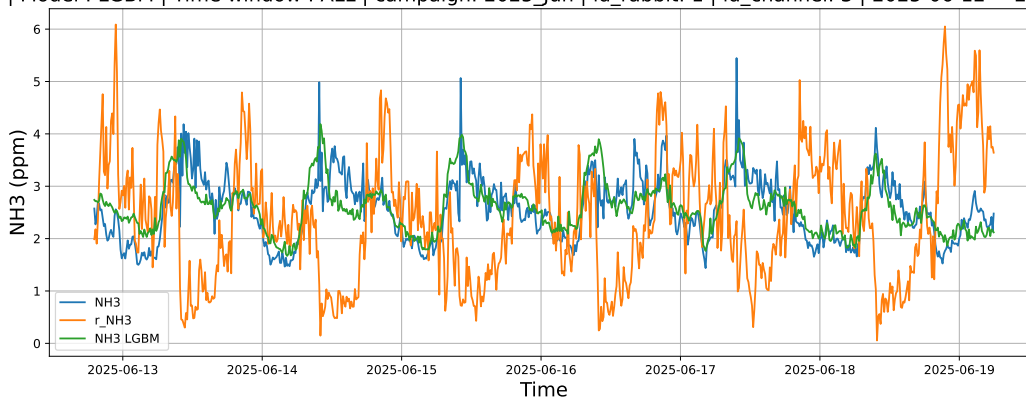


4.1.1.3 Time window : ALL

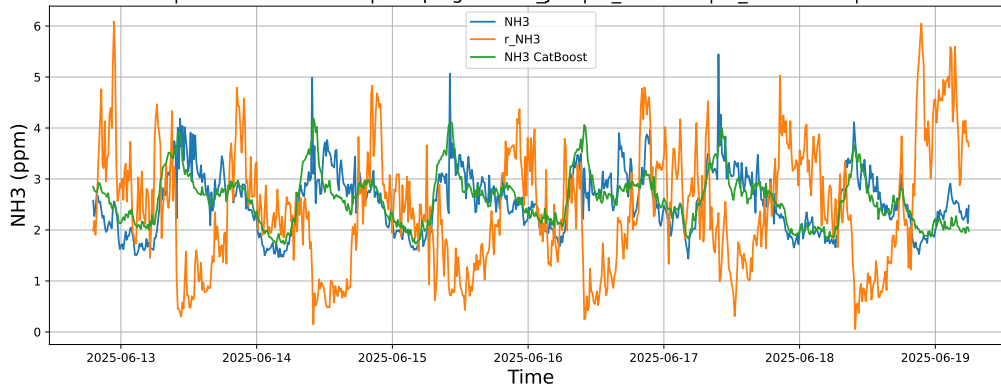
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



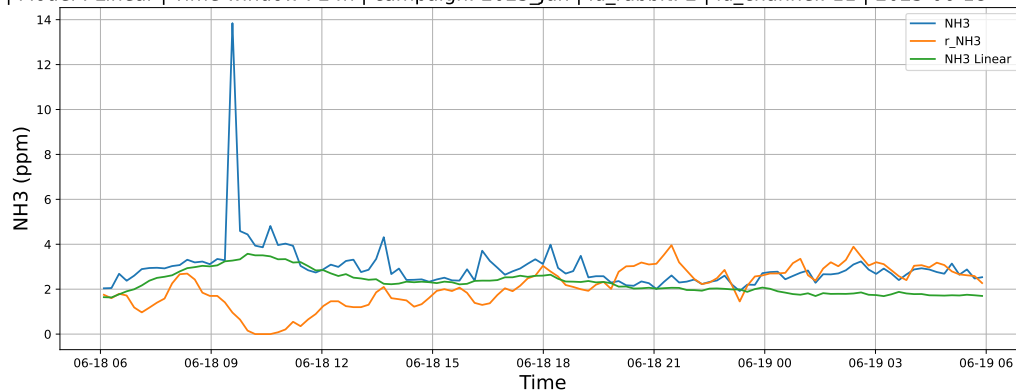
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



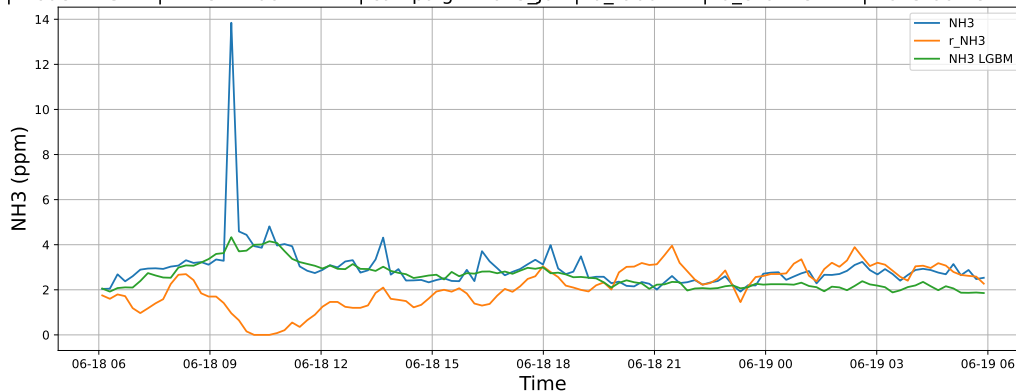
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

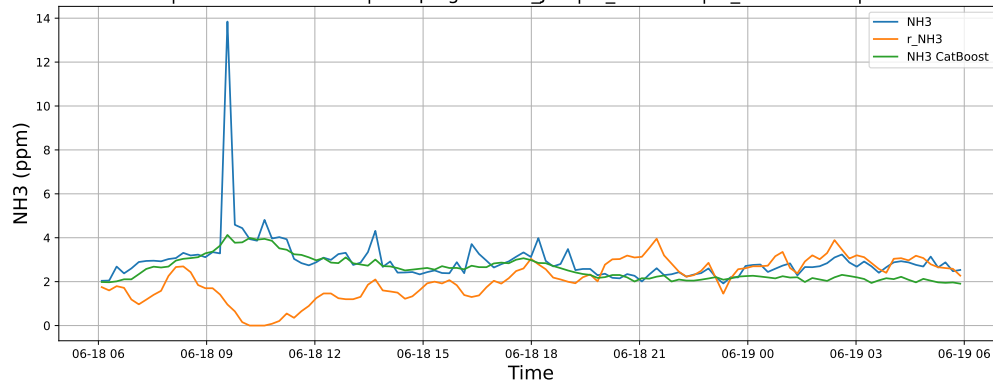
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

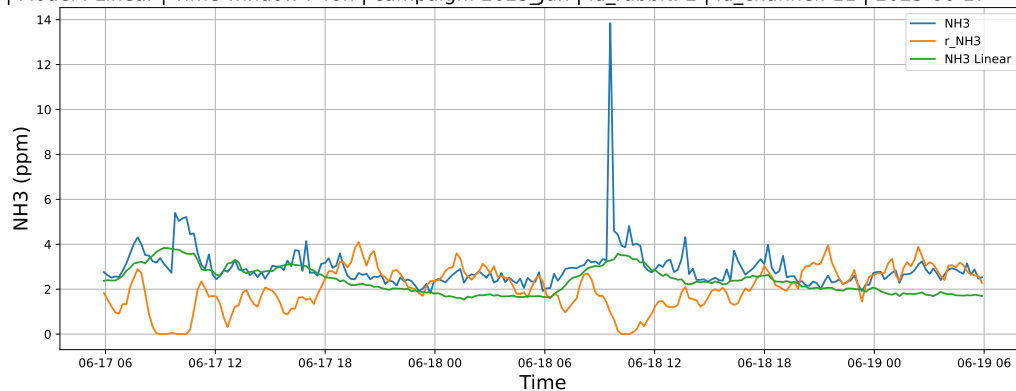


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

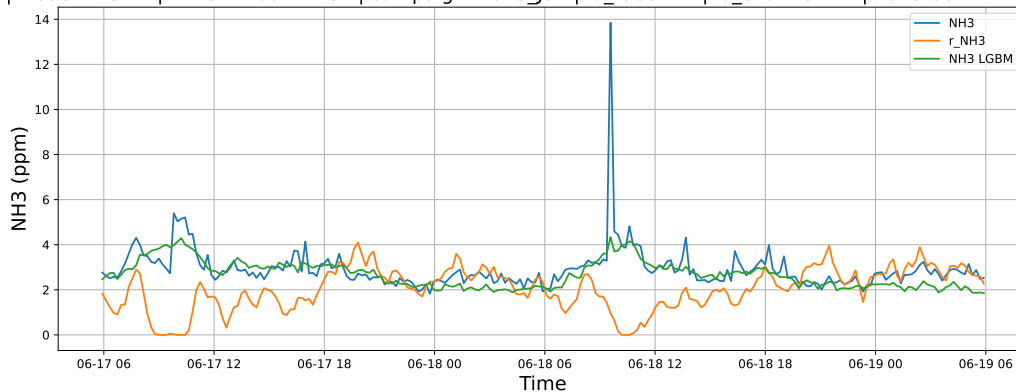


4.1.2.2 Time window : 48

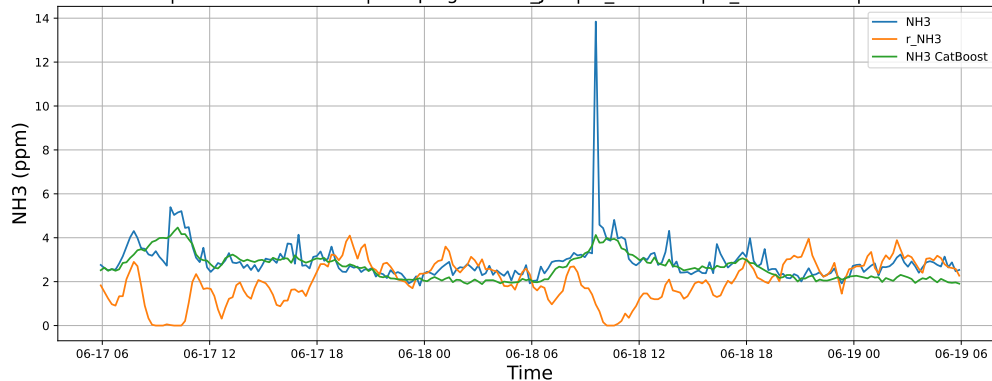
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

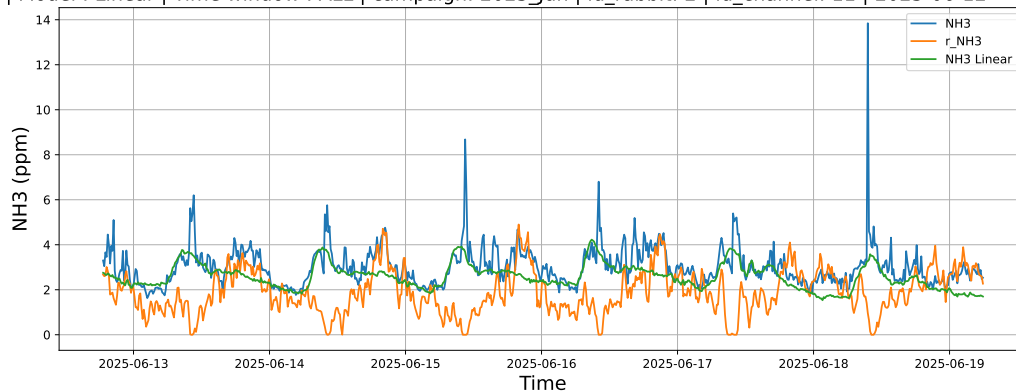


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

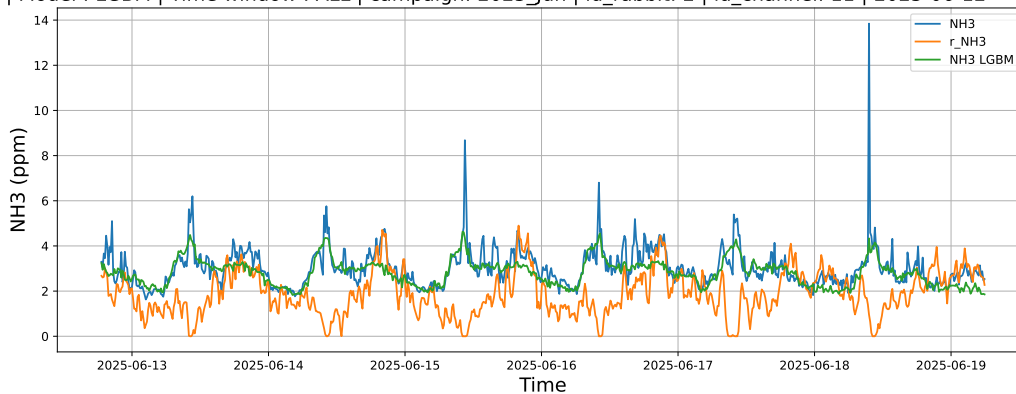


4.1.2.3 Time window : ALL

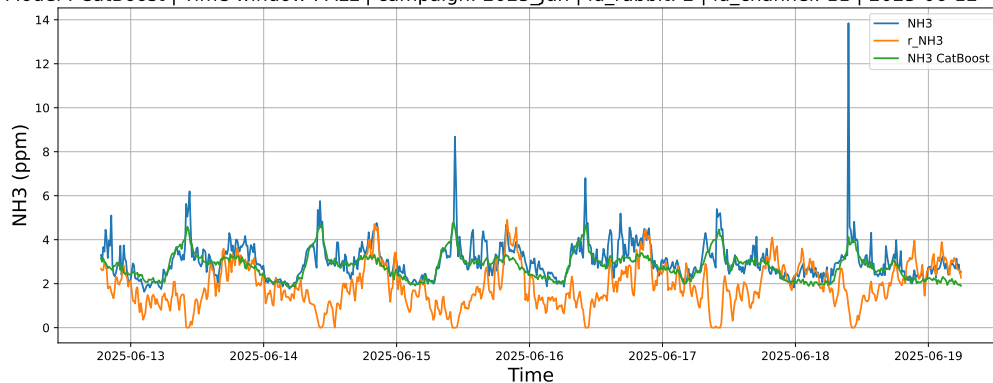
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



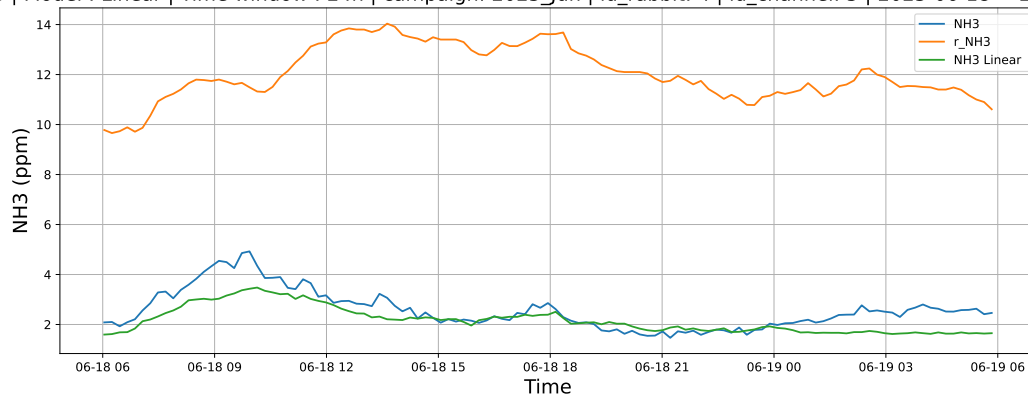
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



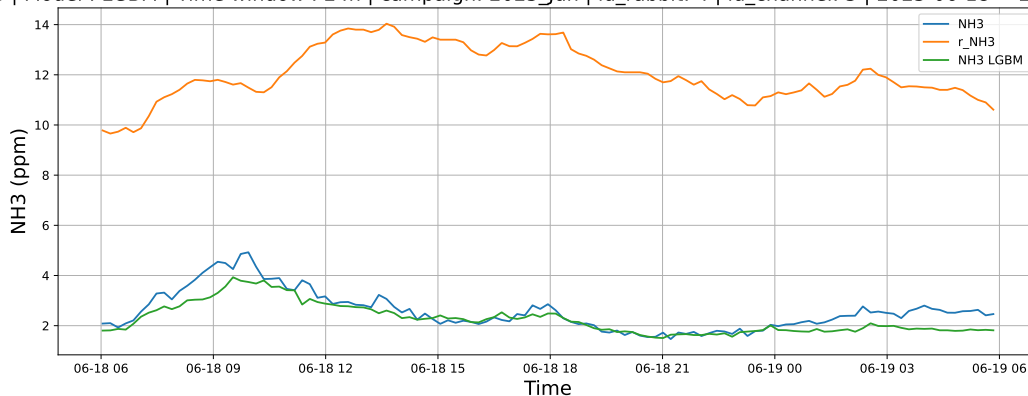
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

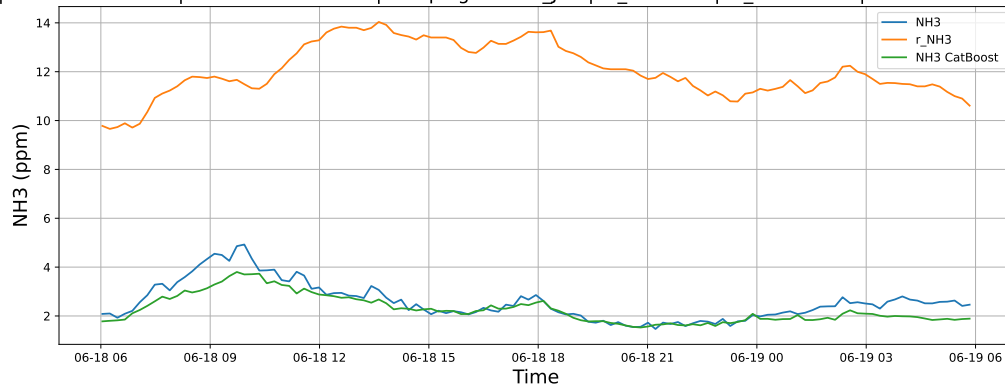
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

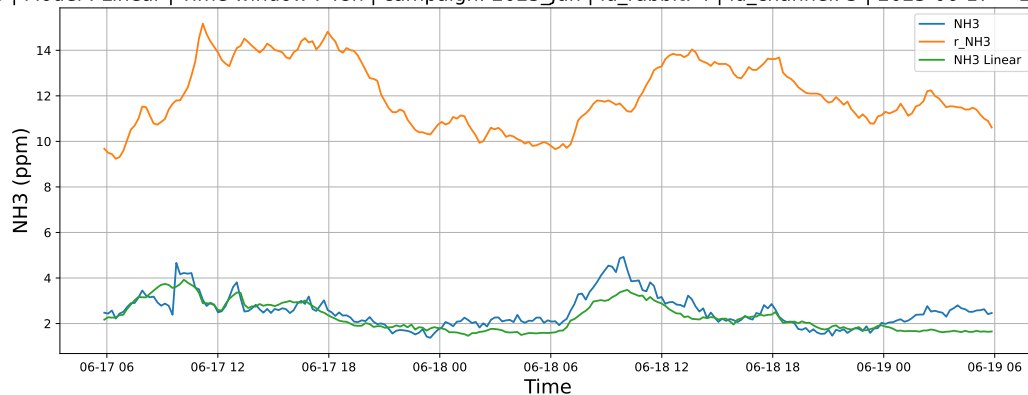


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

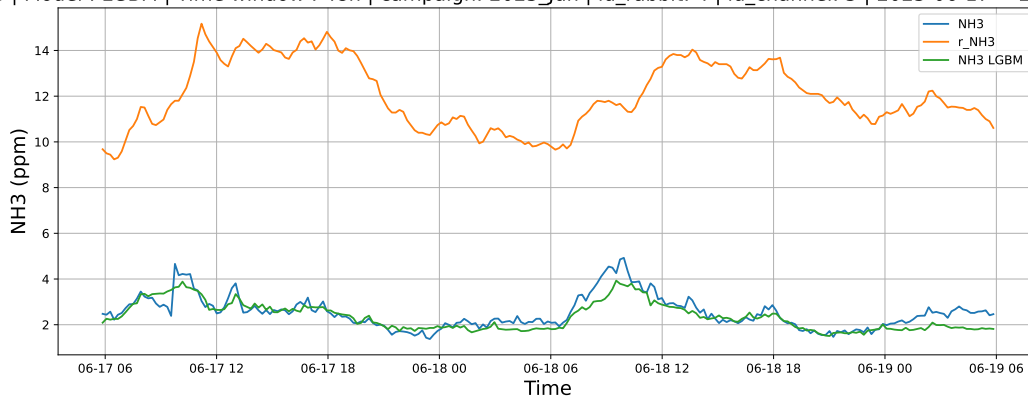


4.1.3.2 Time window : 48

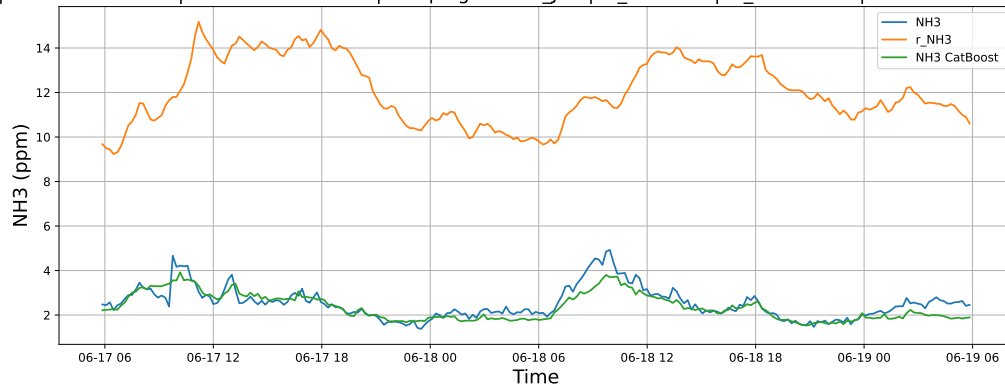
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

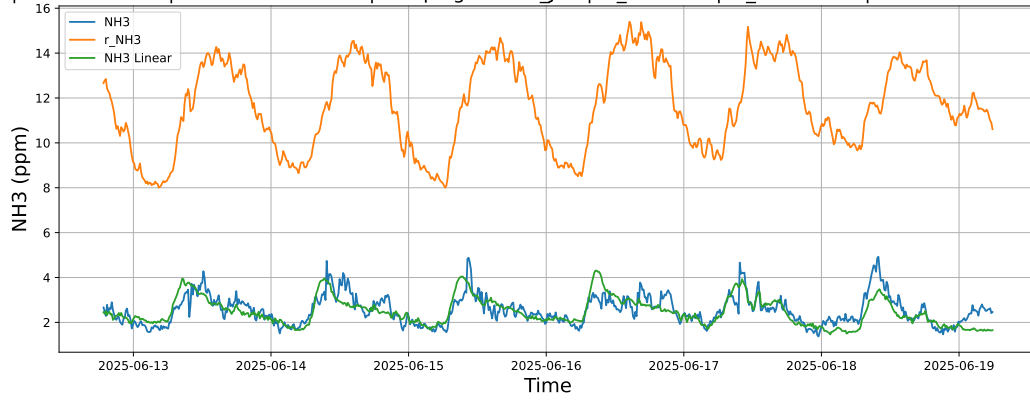


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

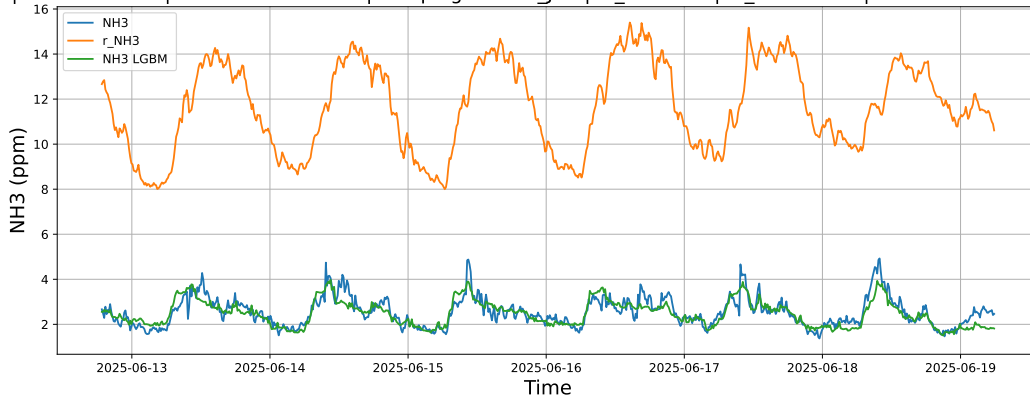


4.1.3.3 Time window : ALL

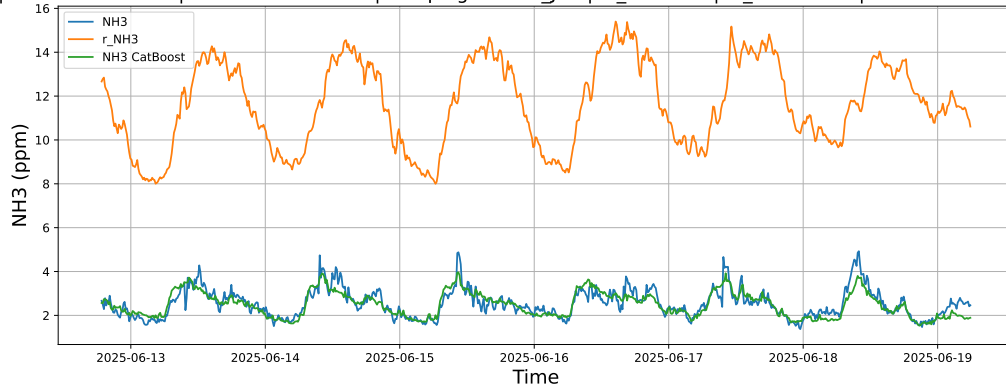
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



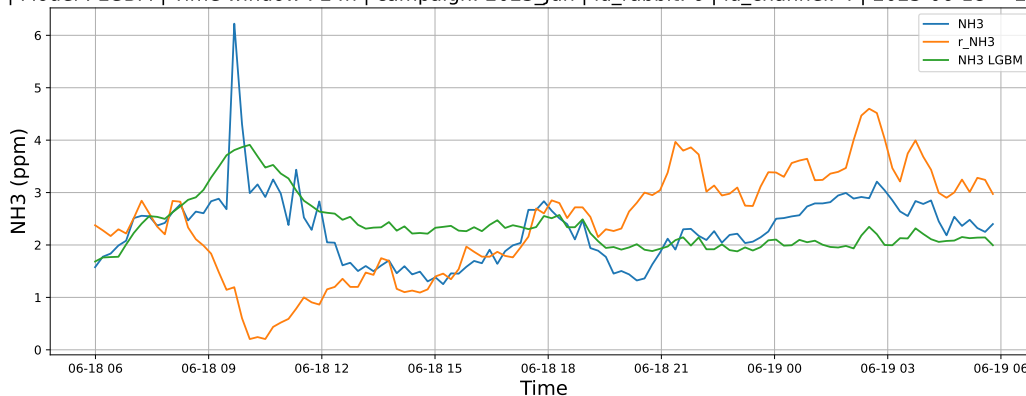
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

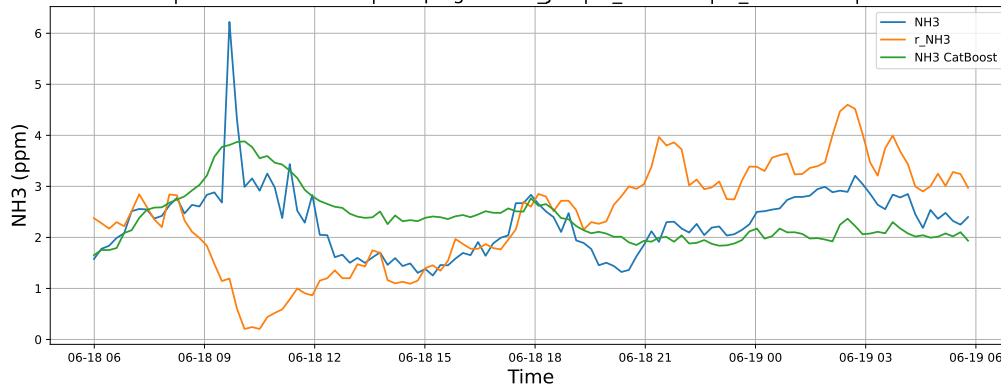
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

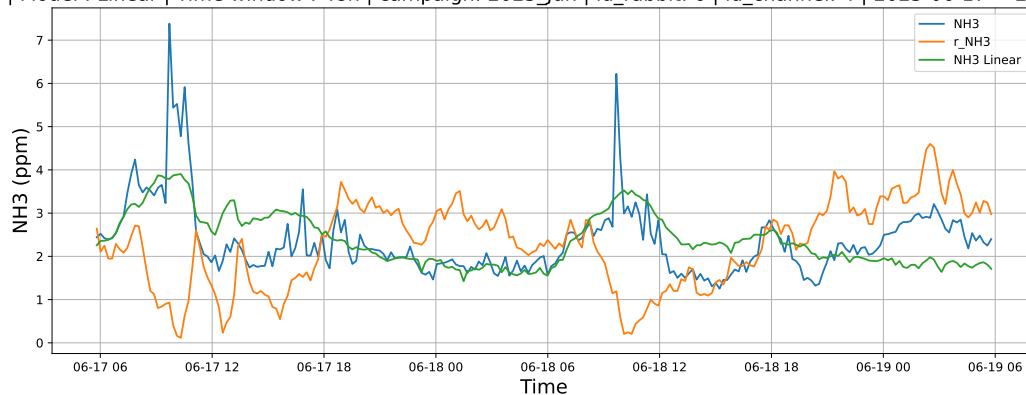


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

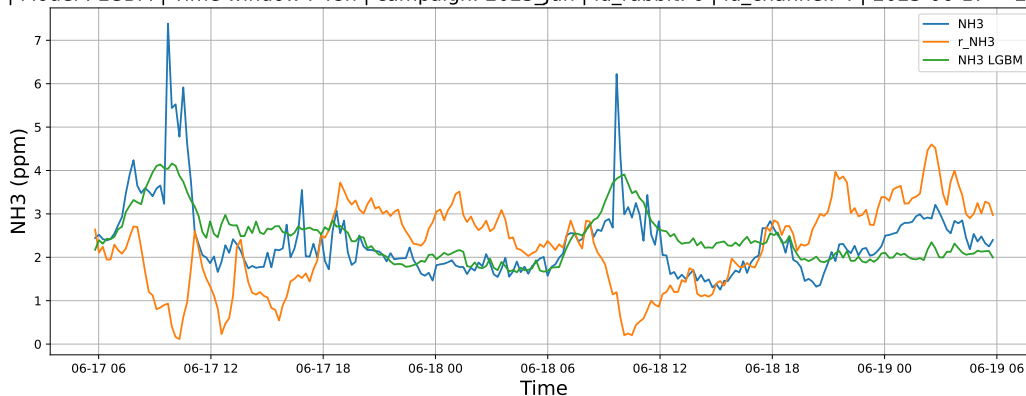


4.1.4.2 Time window : 48

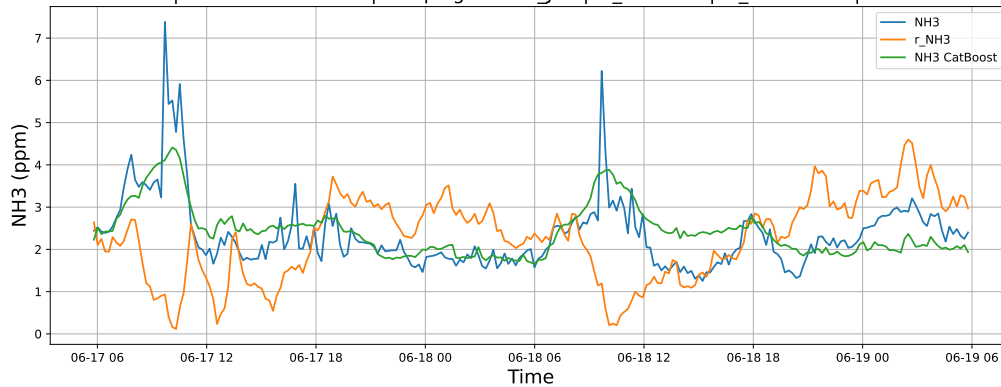
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

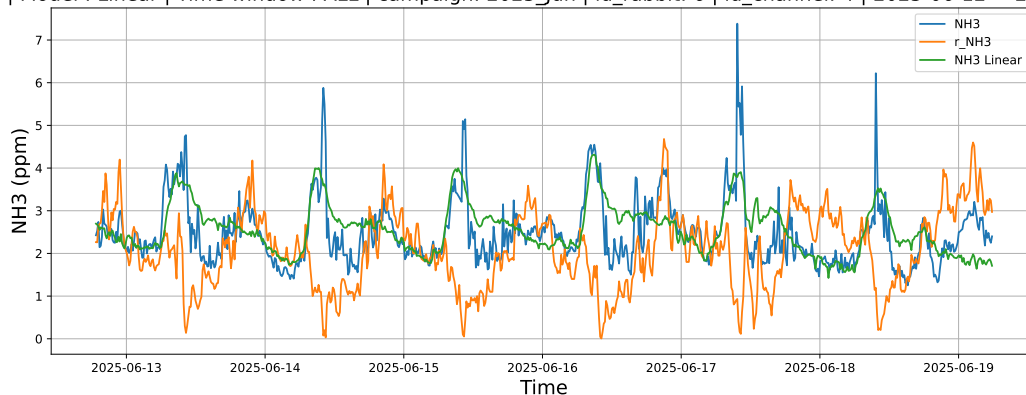


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

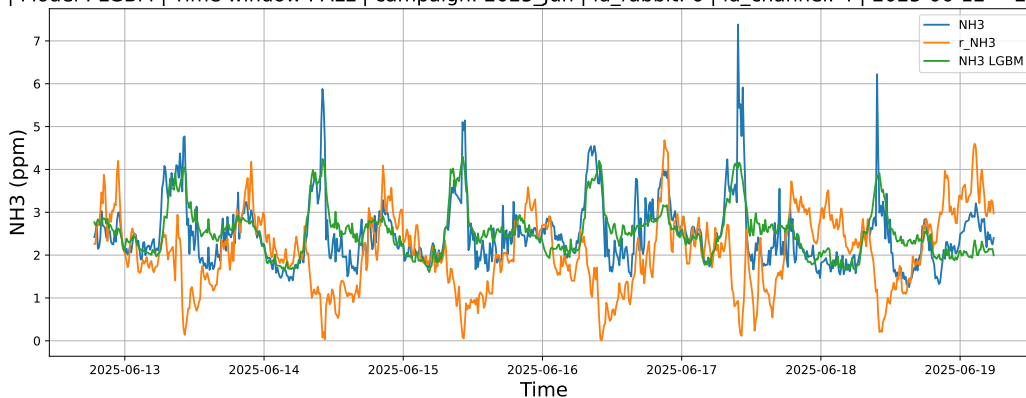


4.1.4.3 Time window : ALL

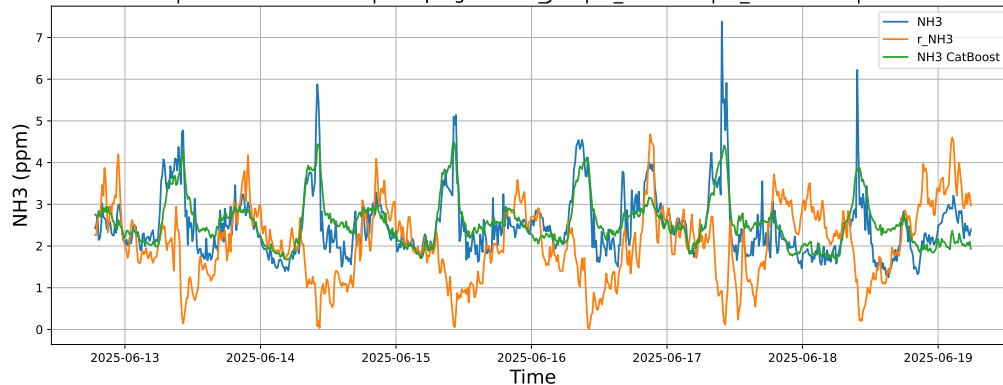
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



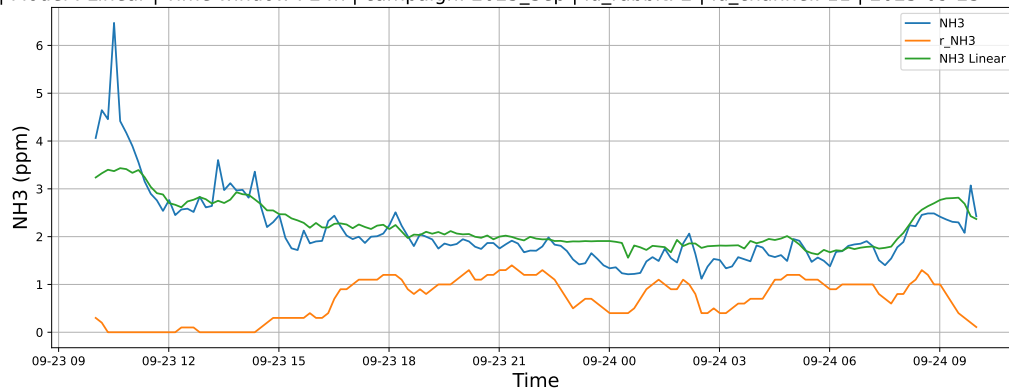
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



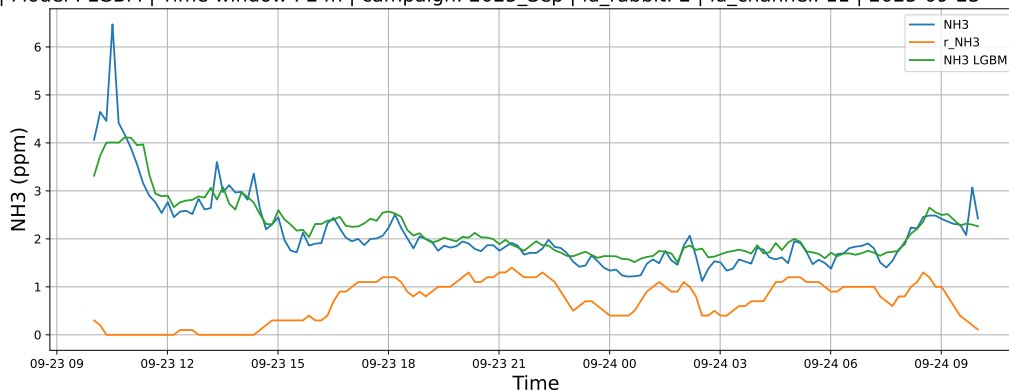
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

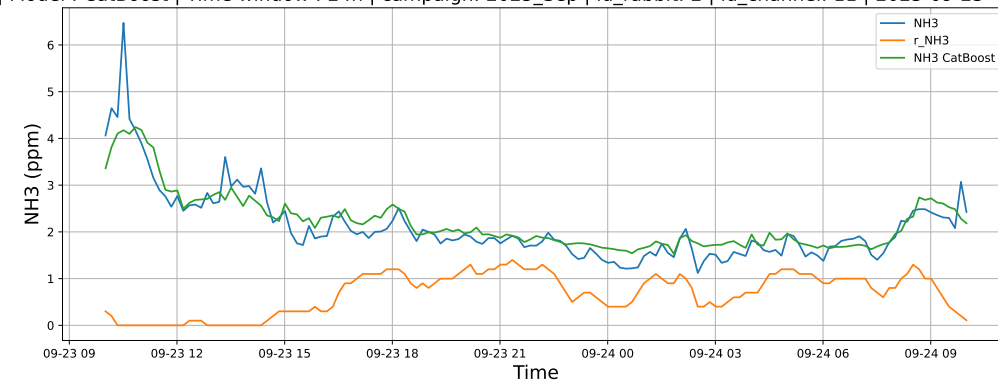
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

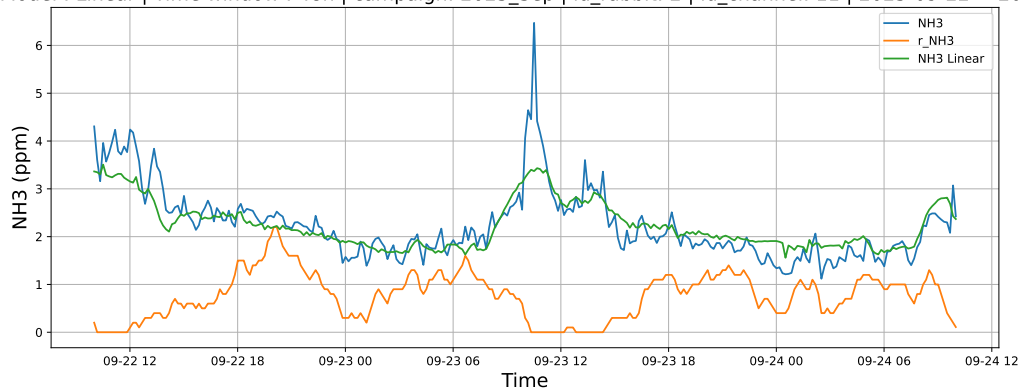


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

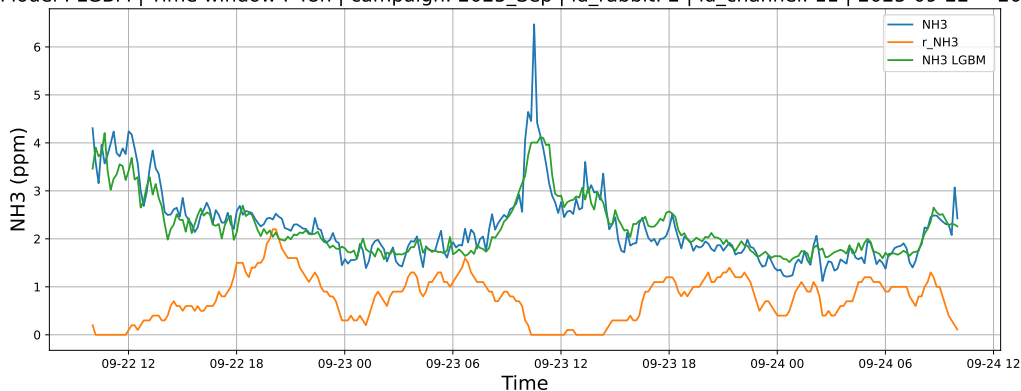


4.1.5.2 Time window : 48

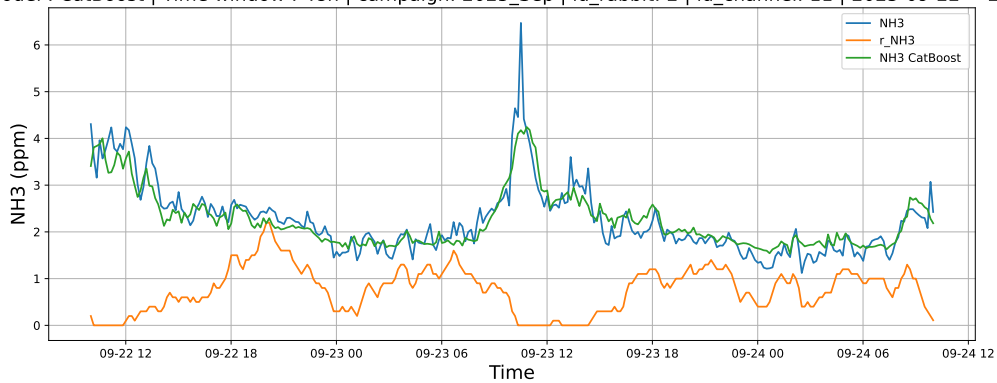
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

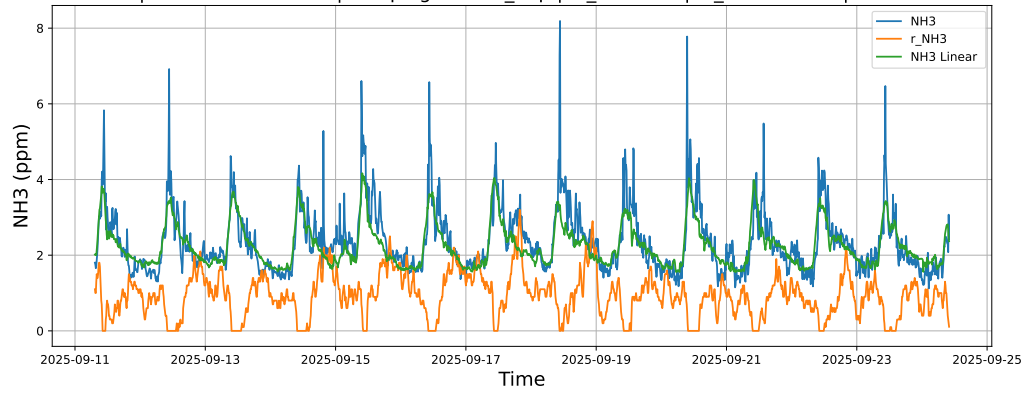


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

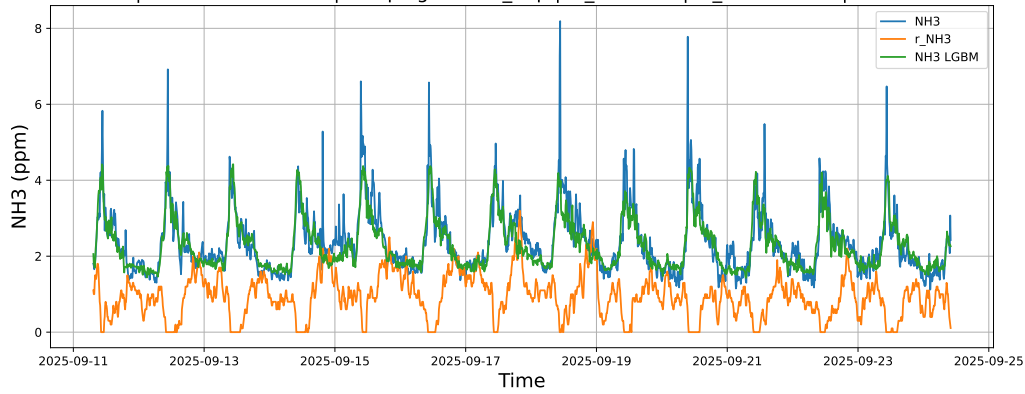


4.1.5.3 Time window : ALL

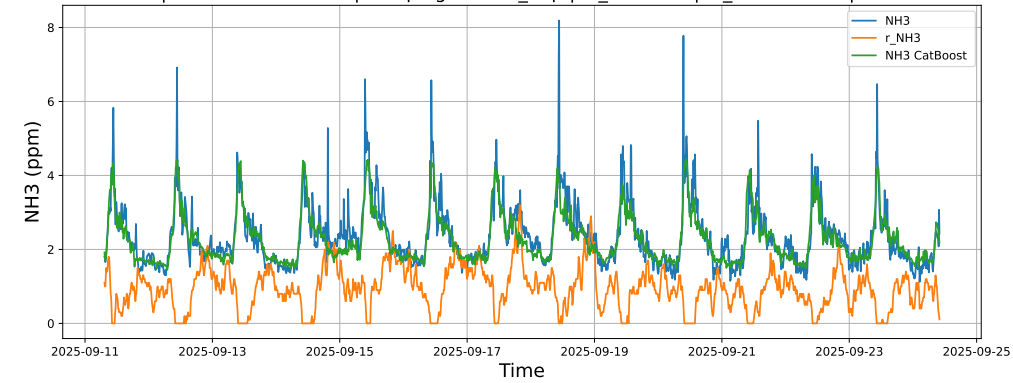
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



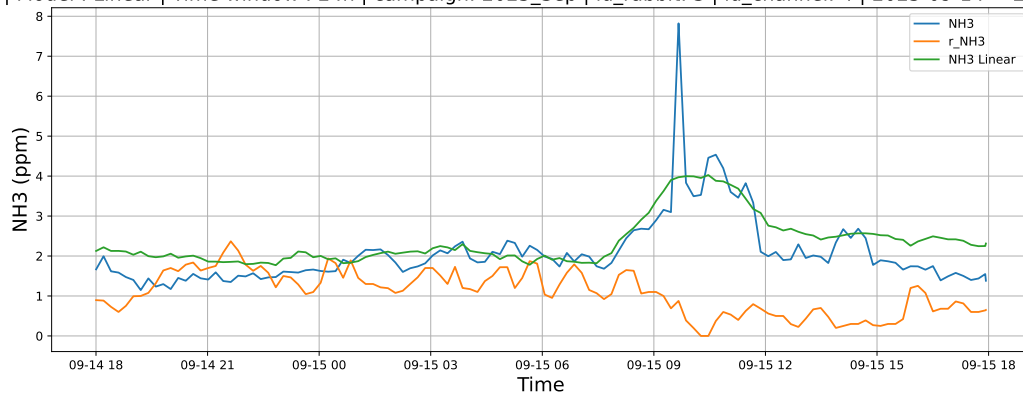
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



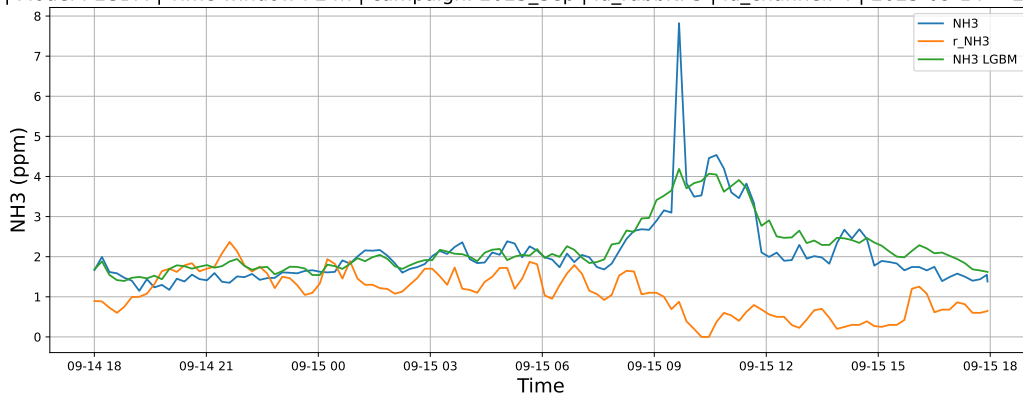
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

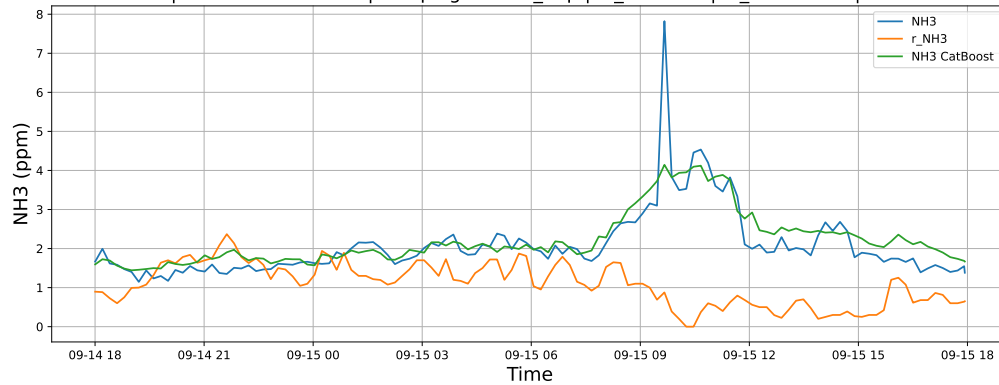
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

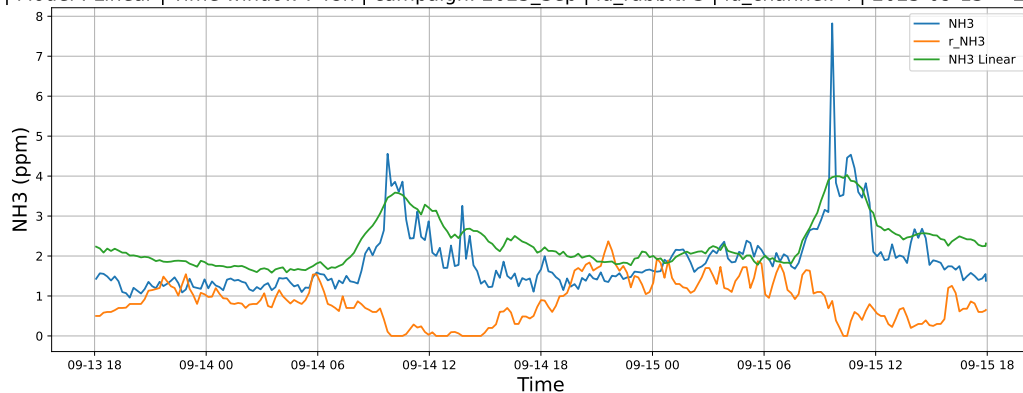


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

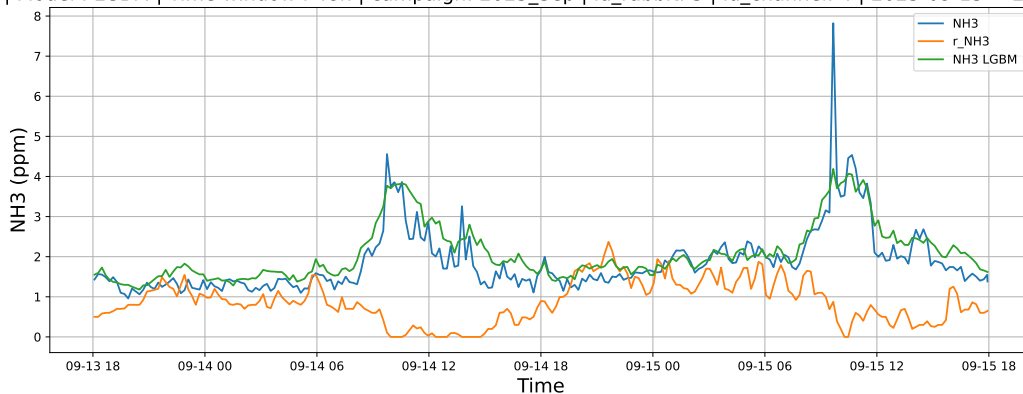


4.1.6.2 Time window : 48

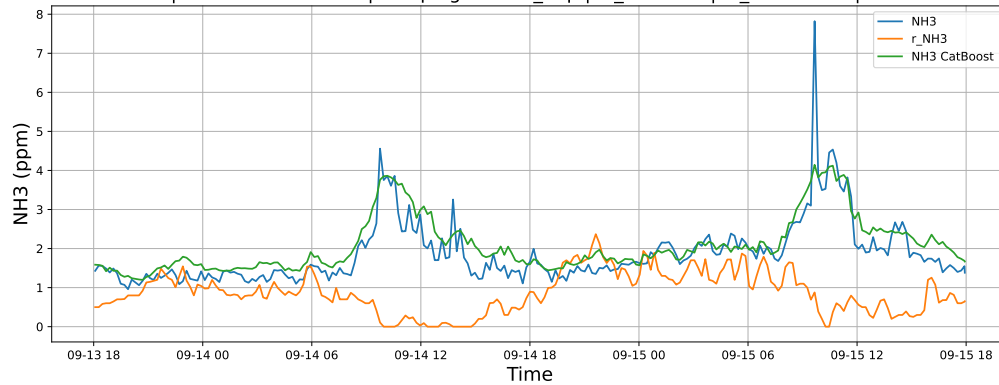
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

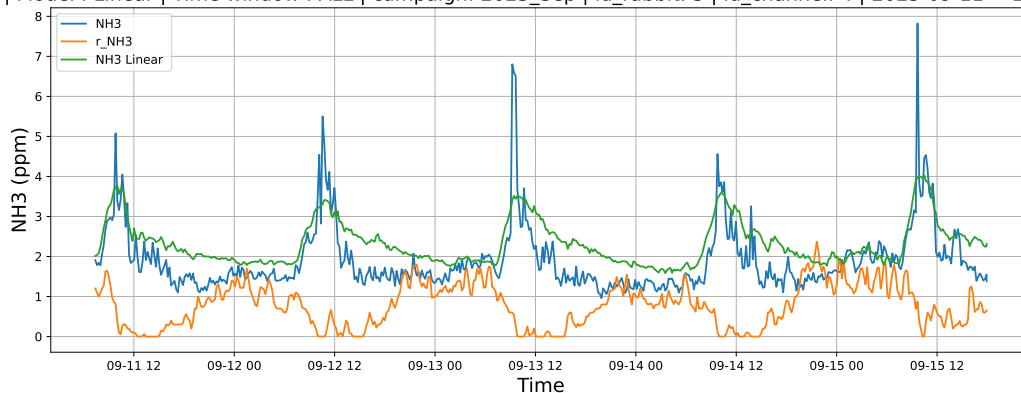


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

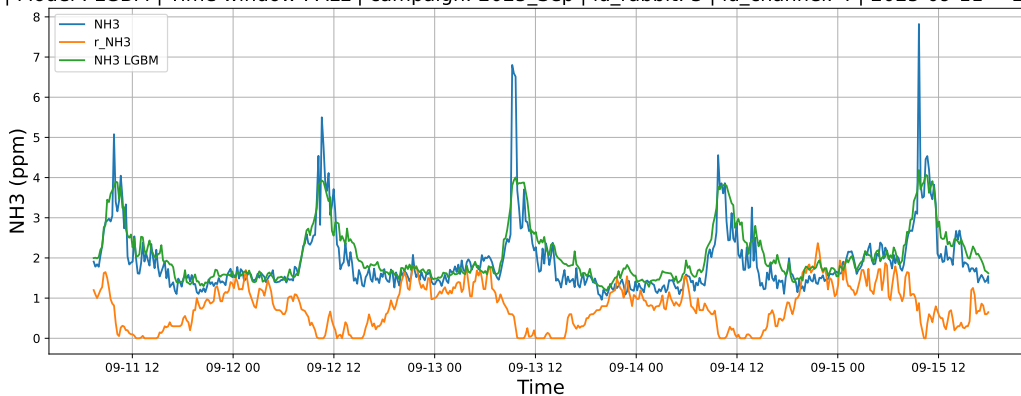


4.1.6.3 Time window : ALL

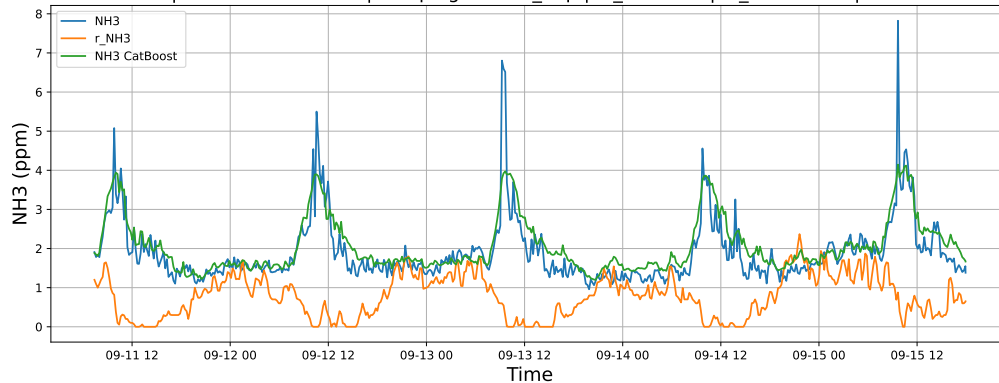
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



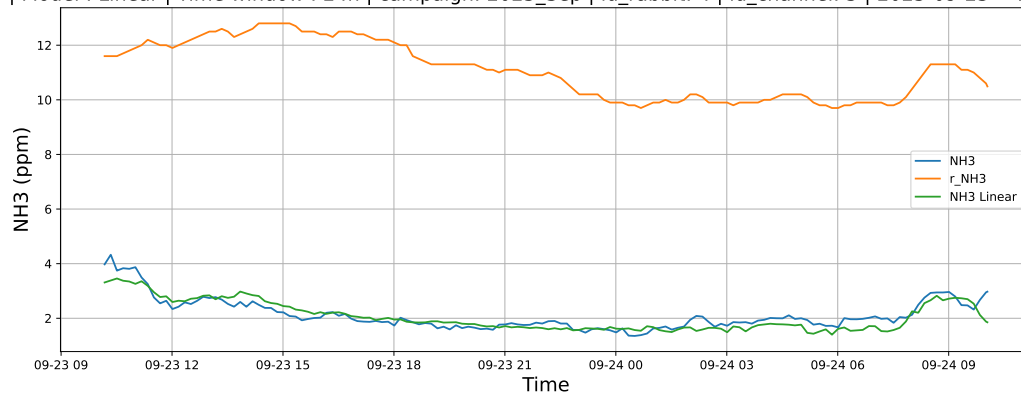
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



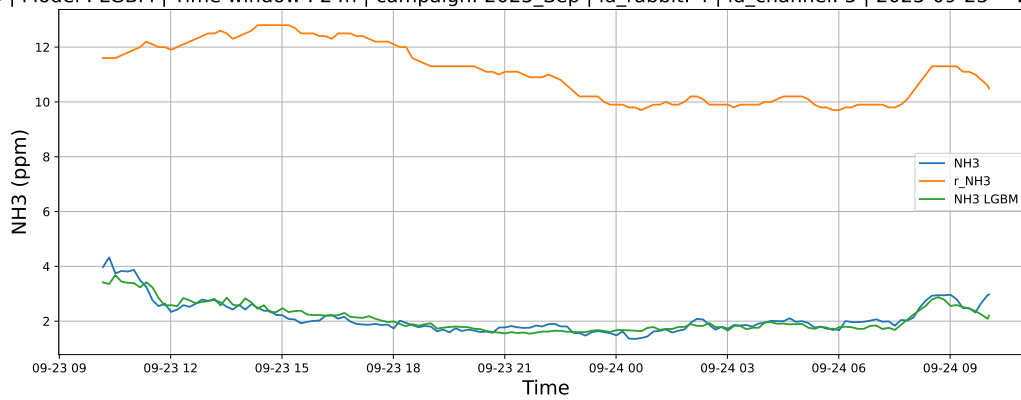
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

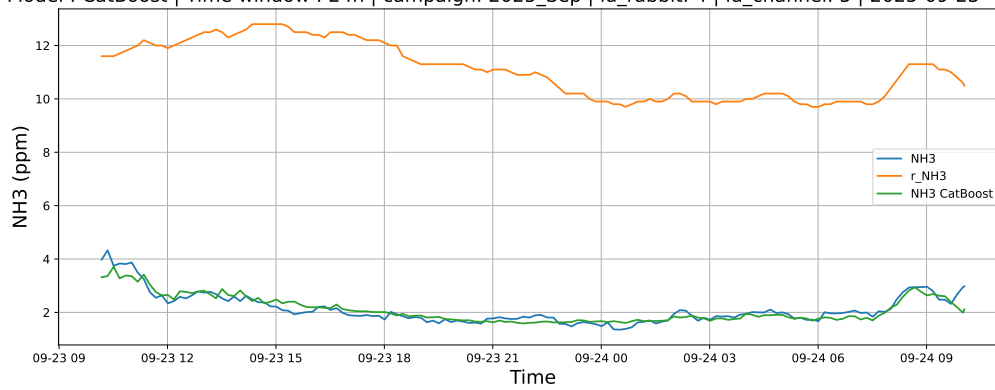
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

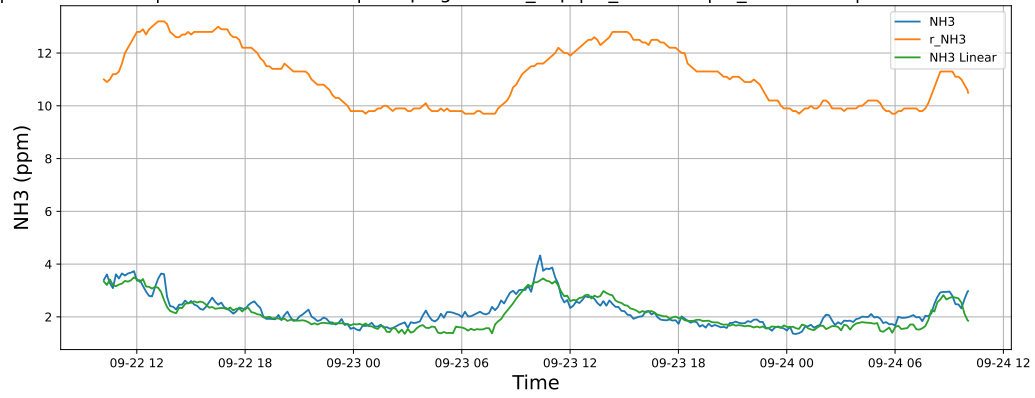


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

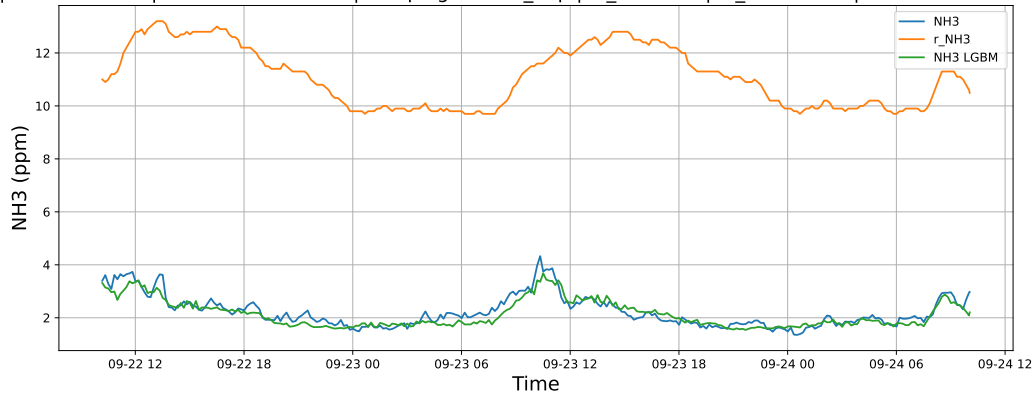


4.1.7.2 Time window : 48

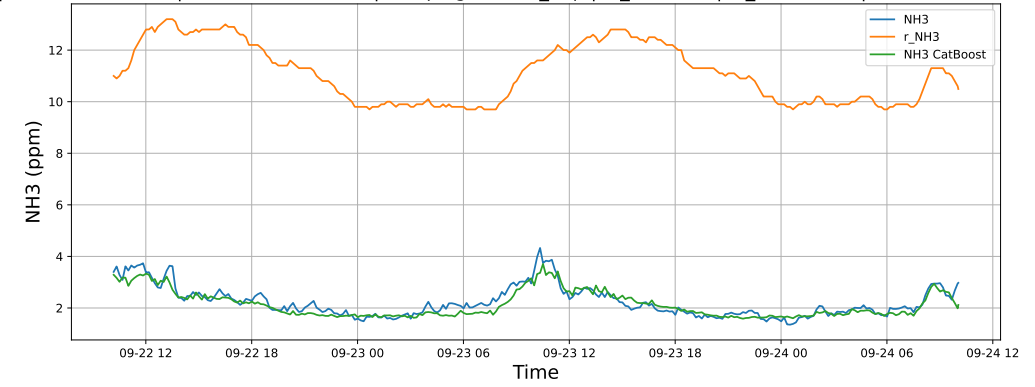
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

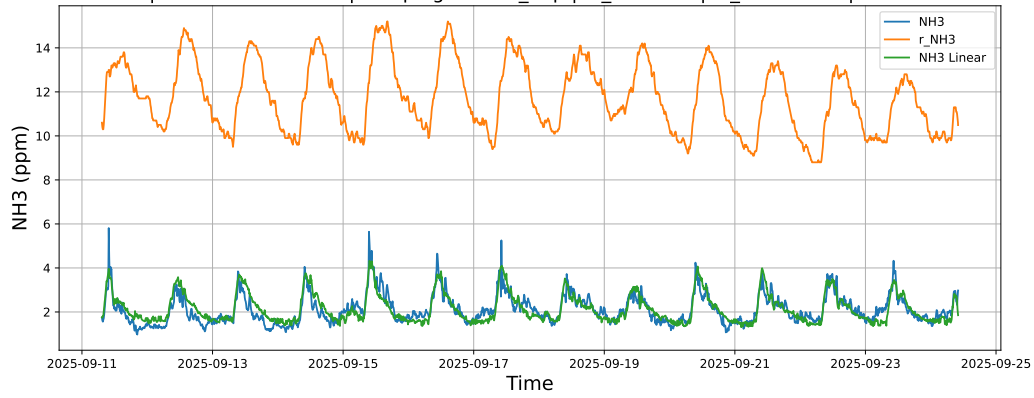


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

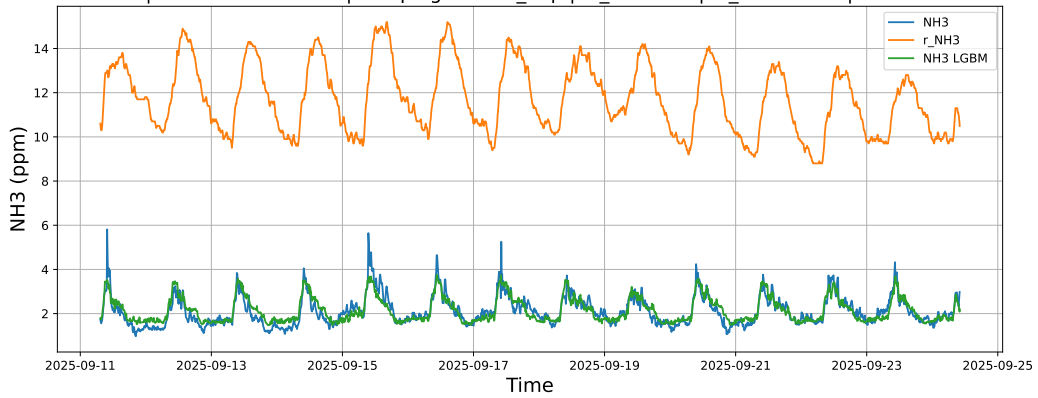


4.1.7.3 Time window : ALL

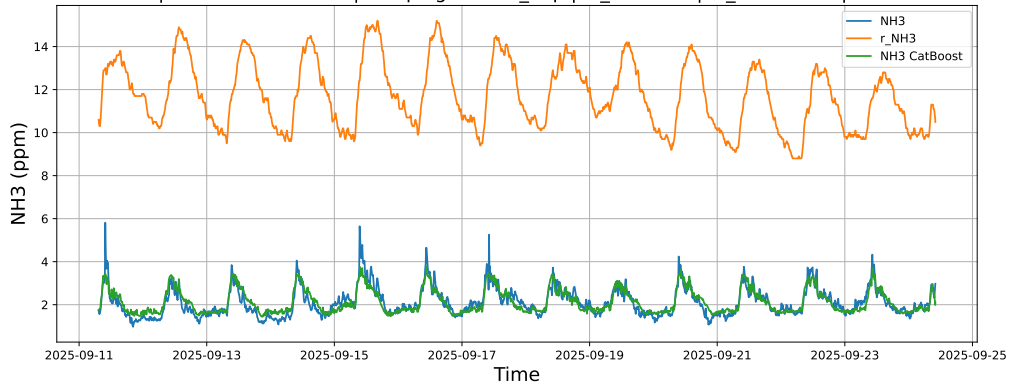
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



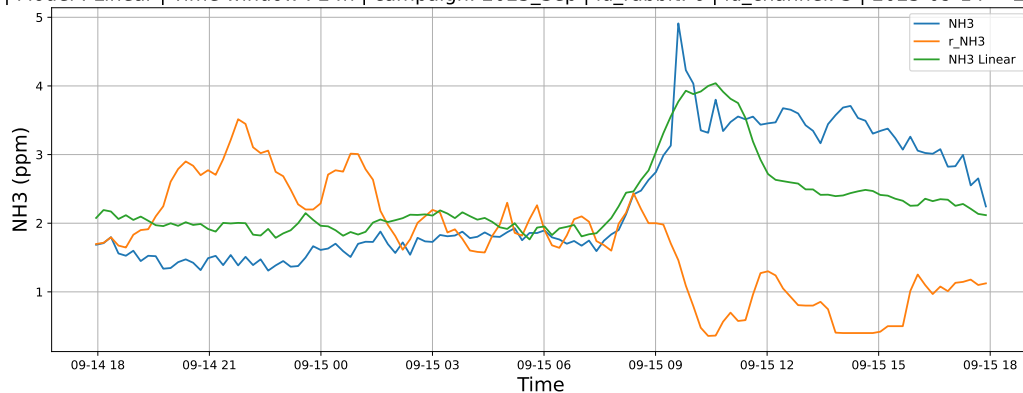
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



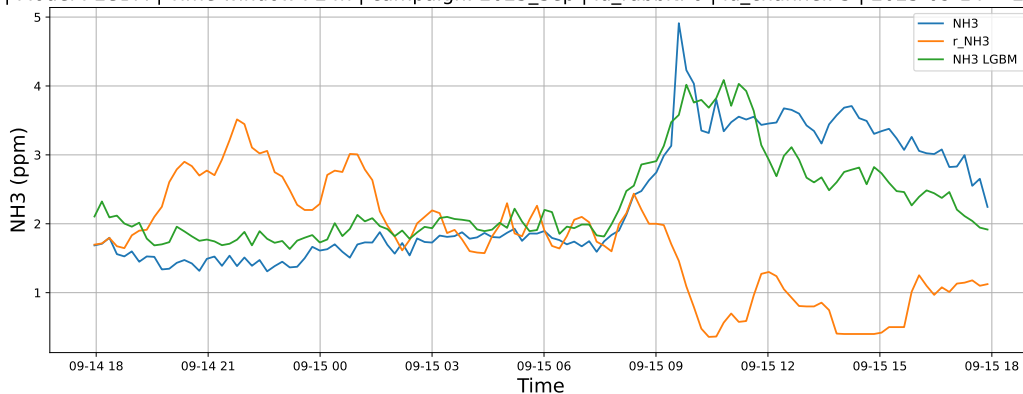
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

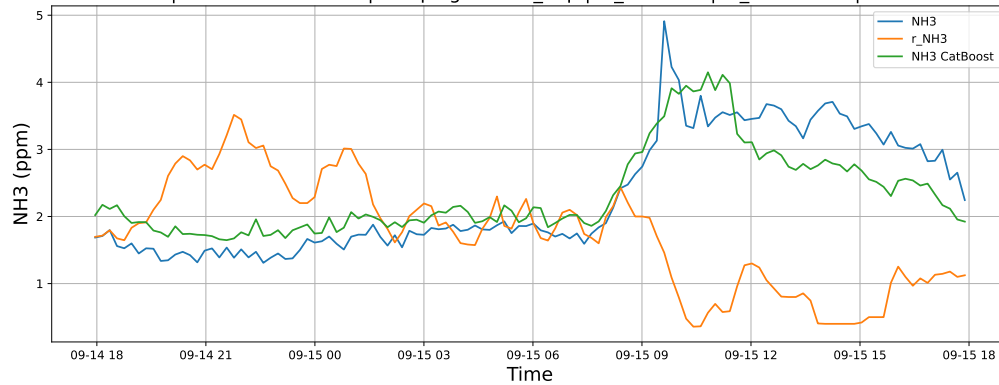
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

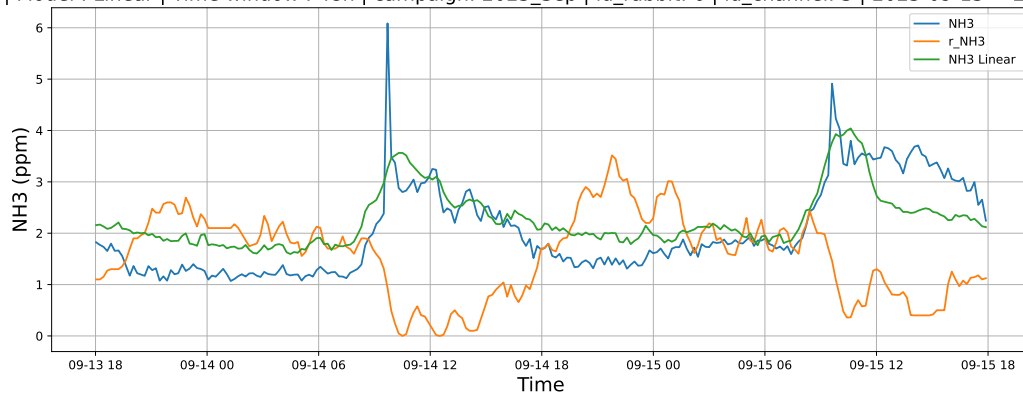


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

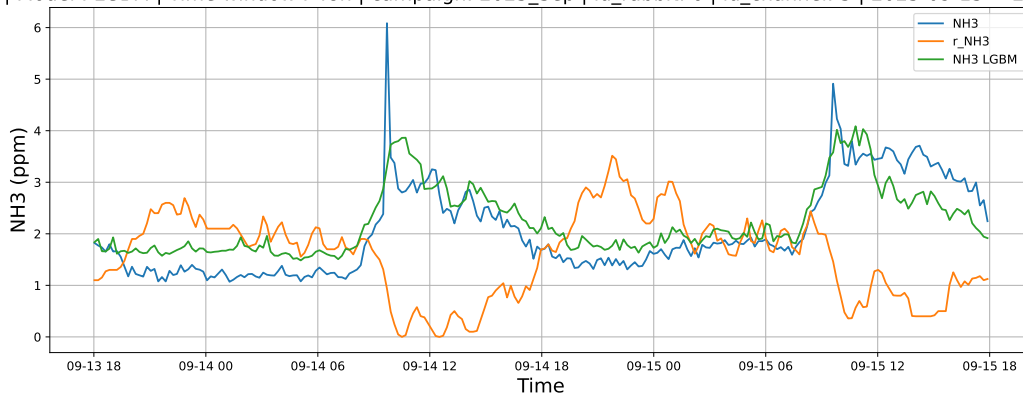


4.1.8.2 Time window : 48

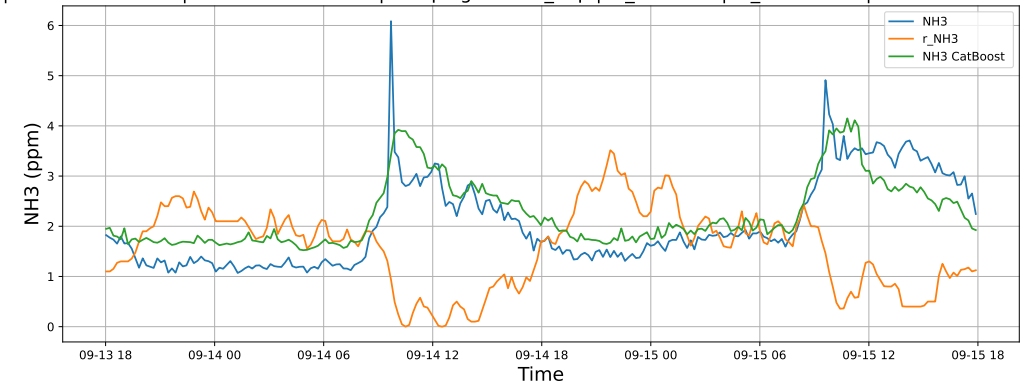
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

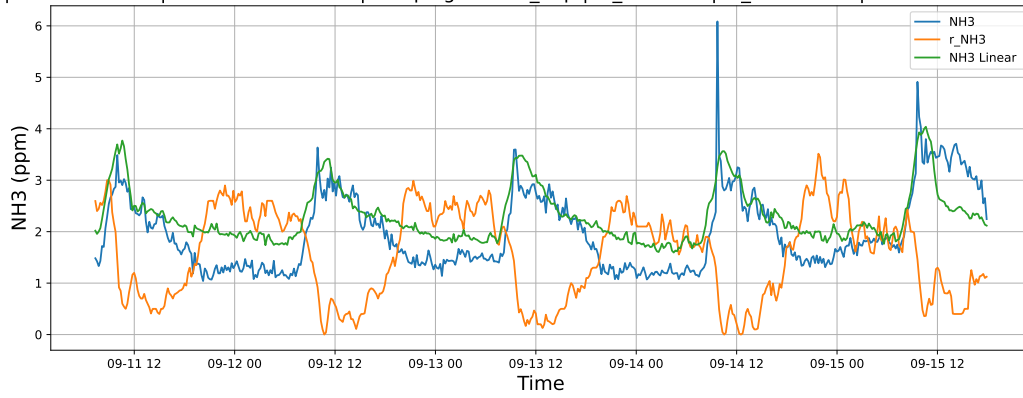


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

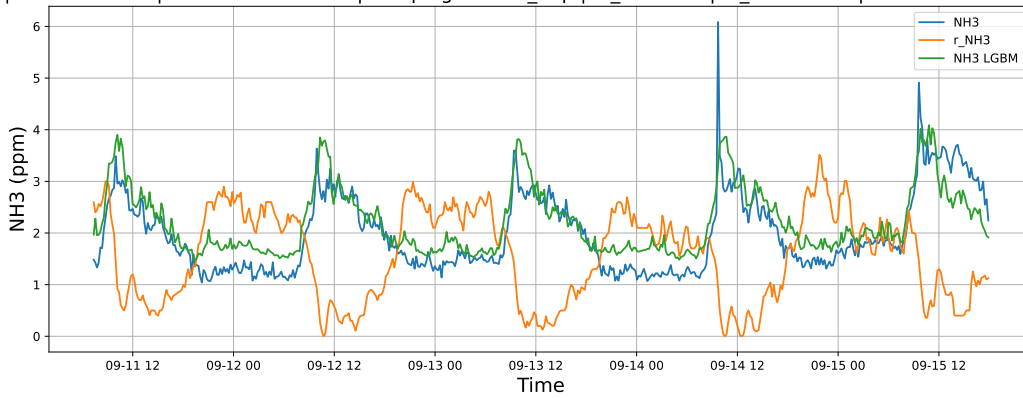


4.1.8.3 Time window : ALL

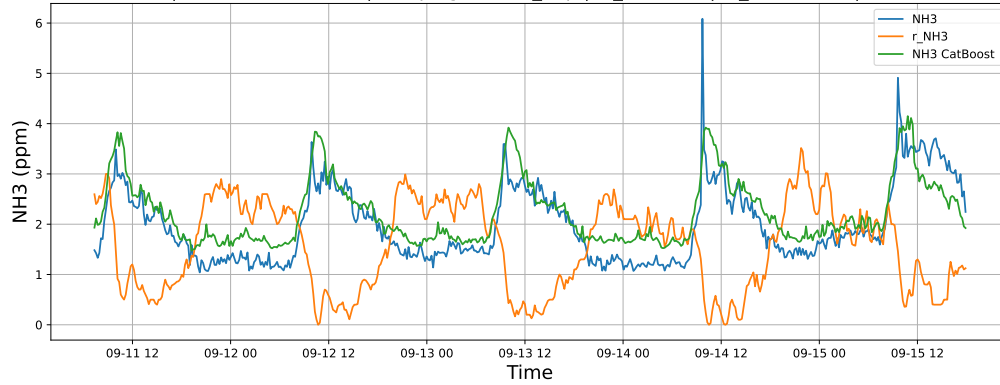
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

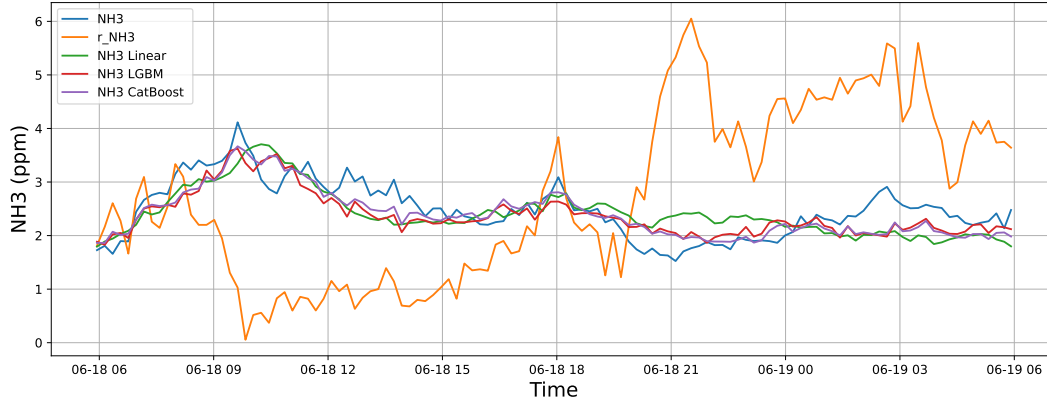


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

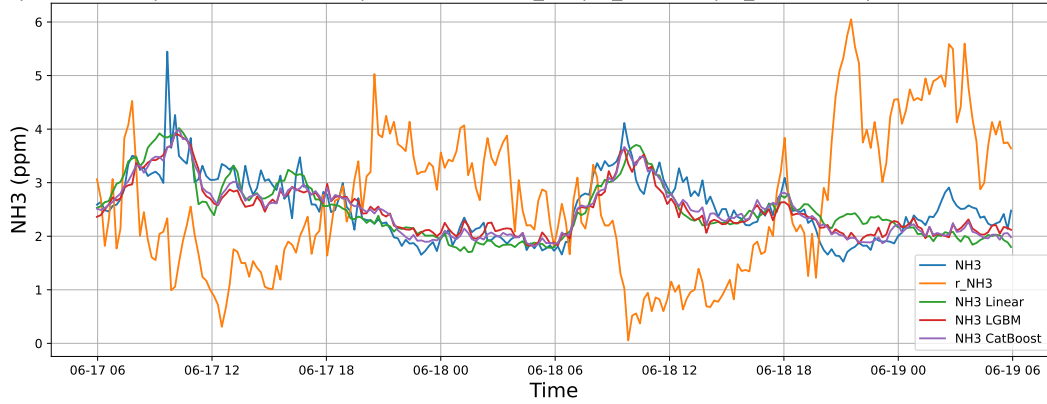
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



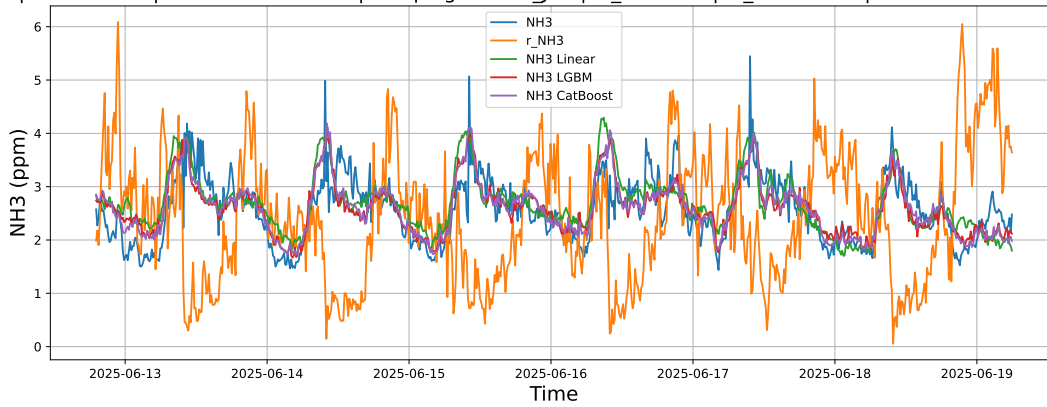
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

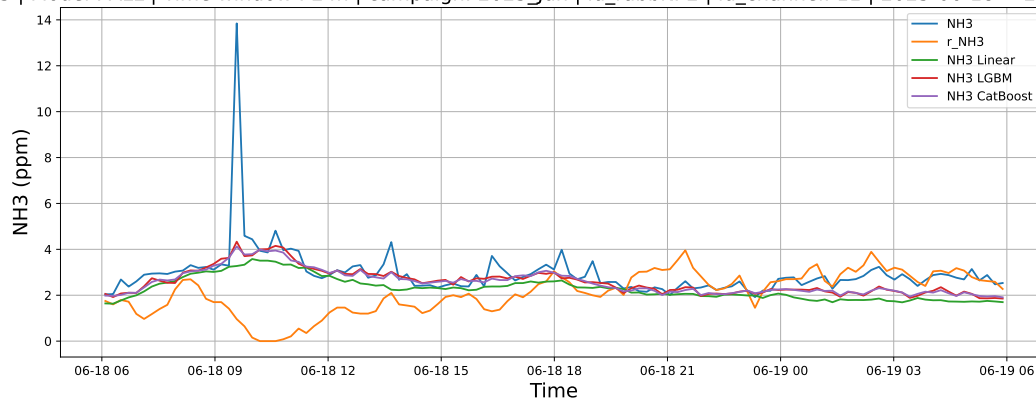
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

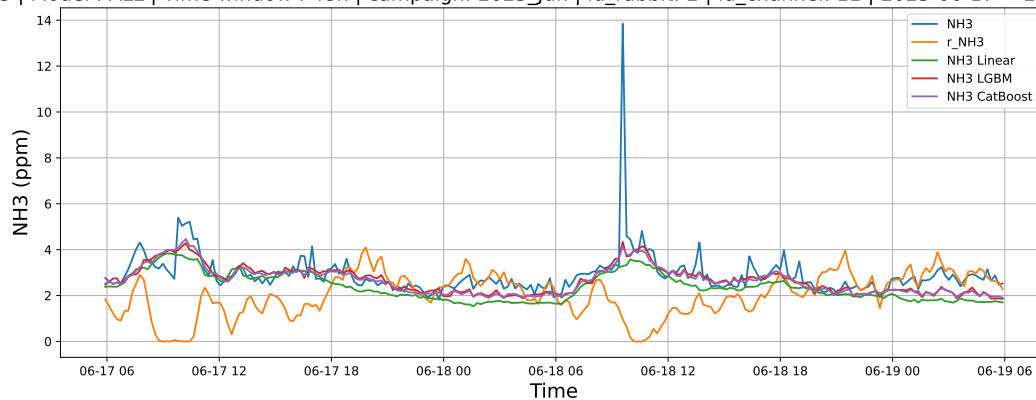
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



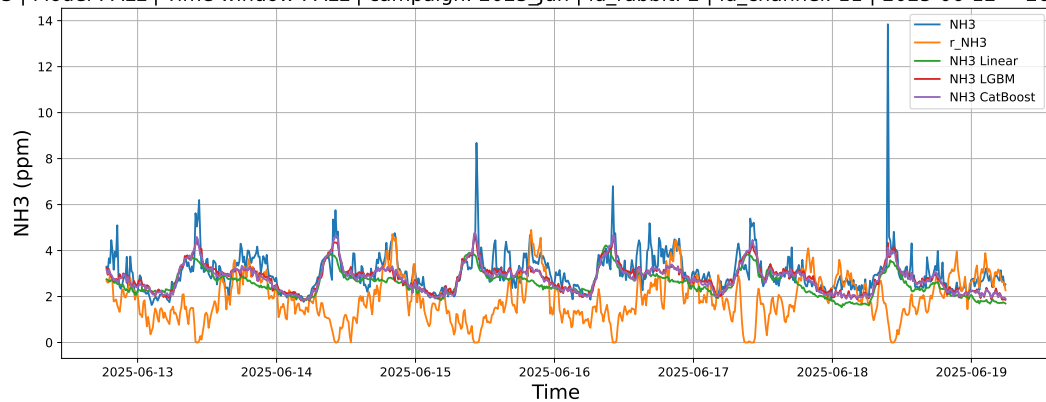
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

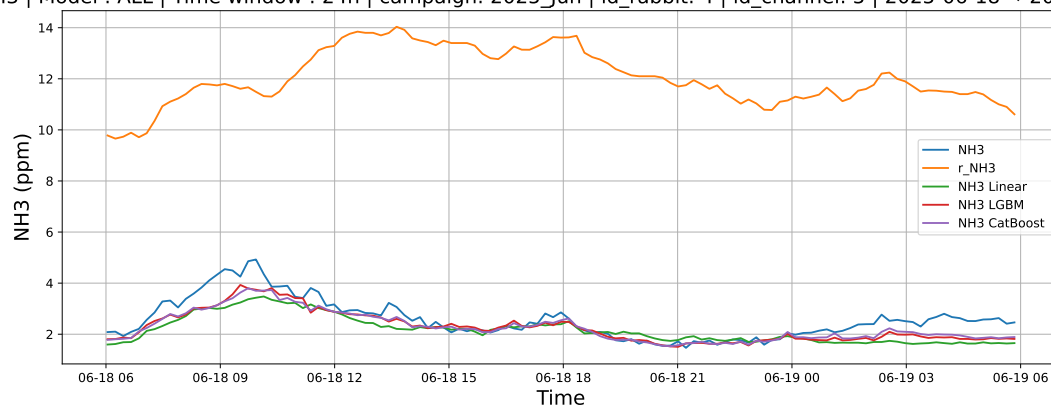
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

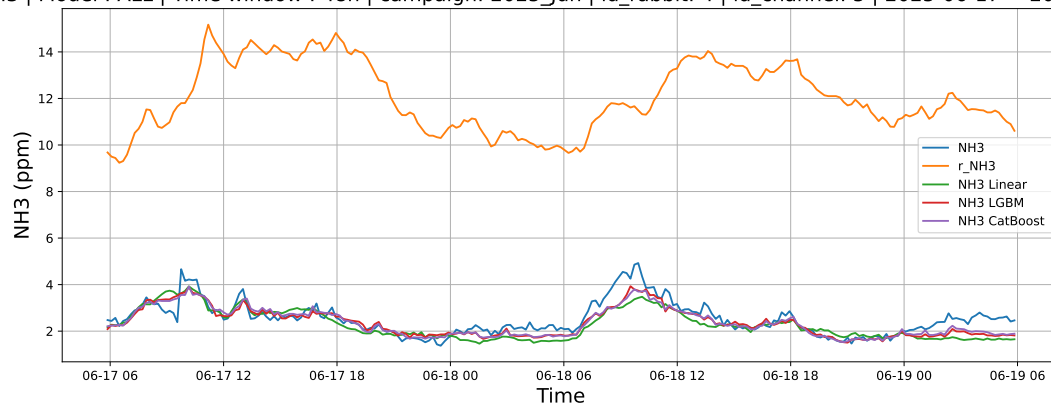
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



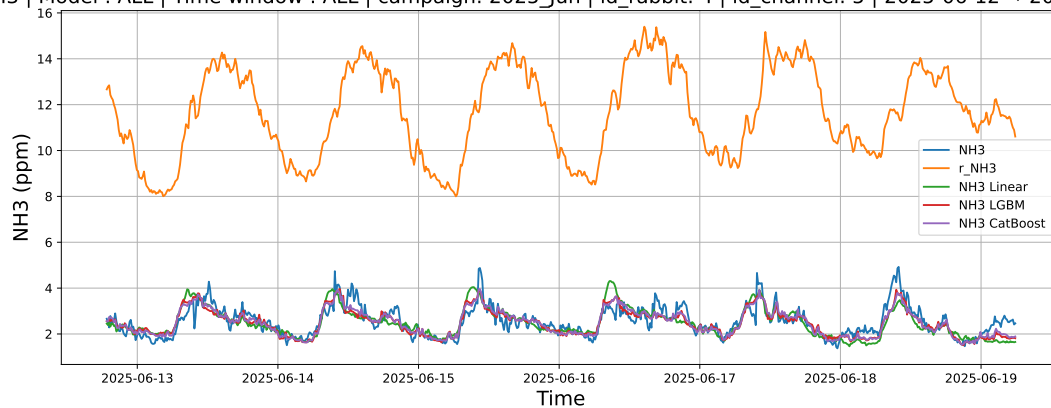
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

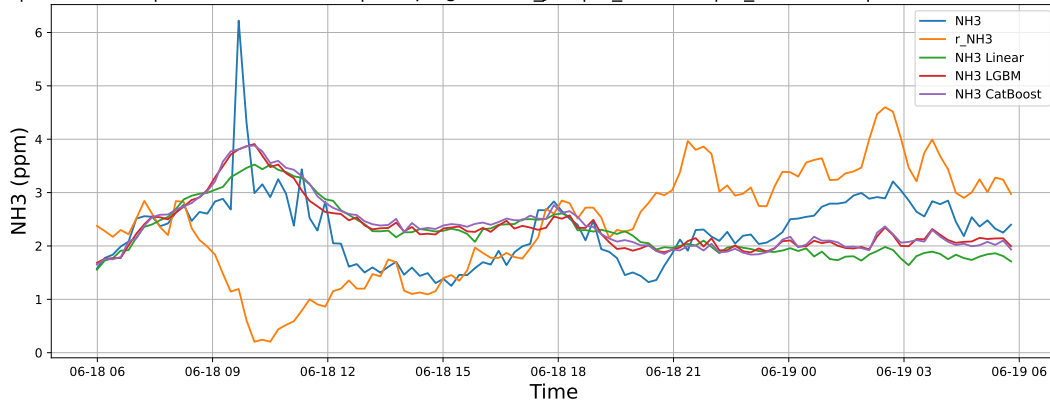
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

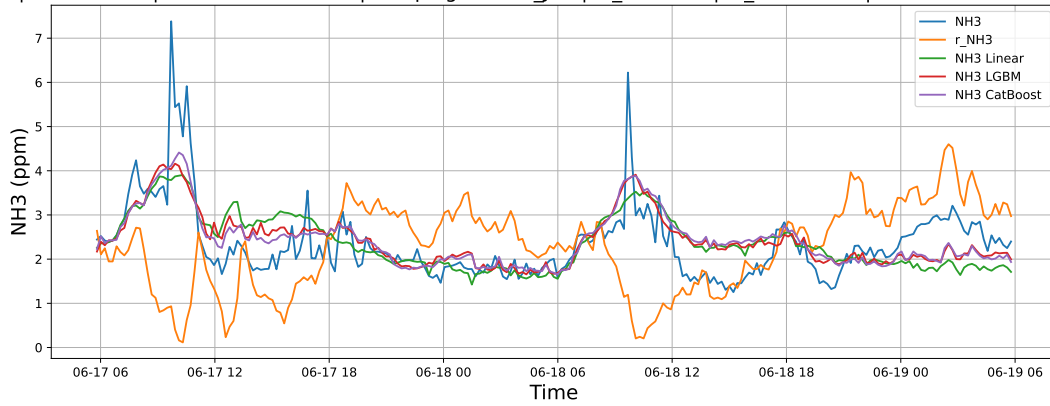
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



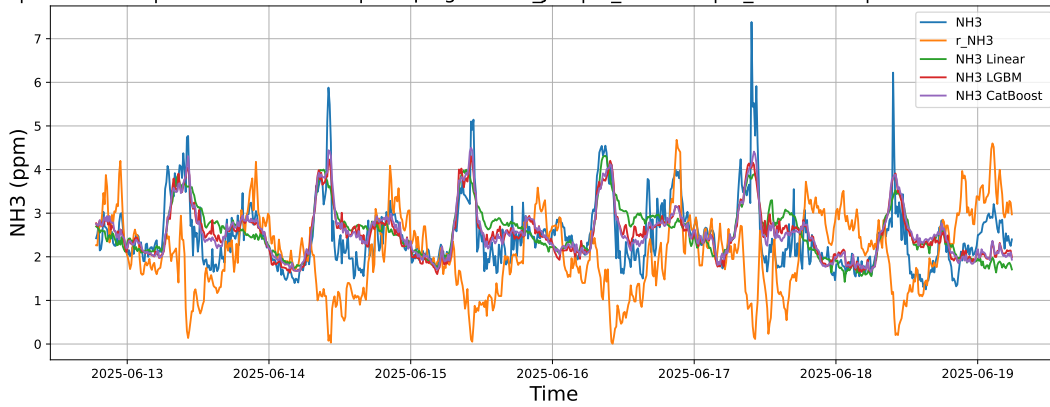
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

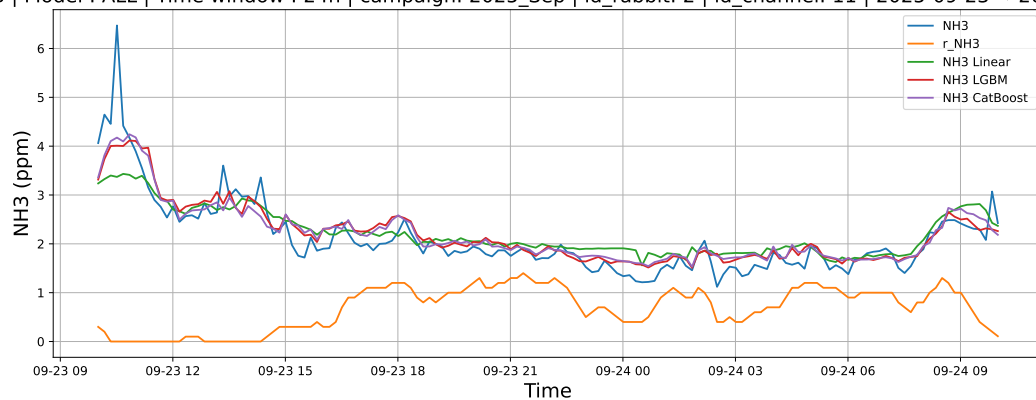
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

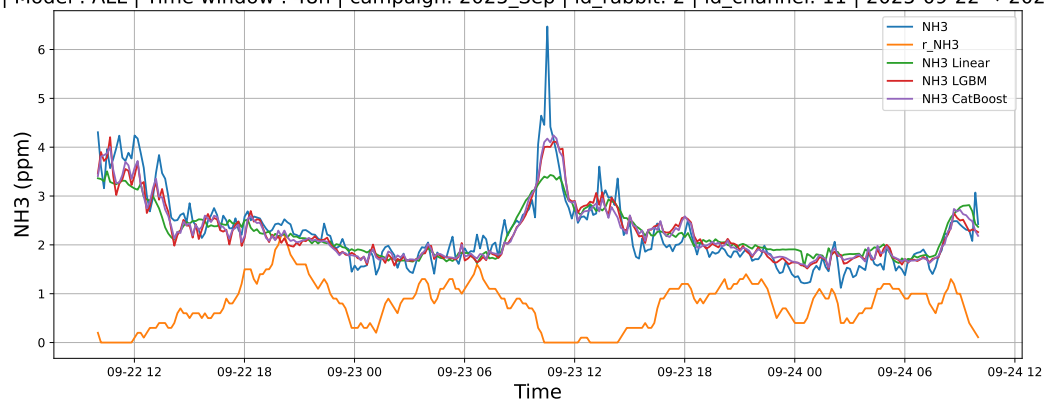
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



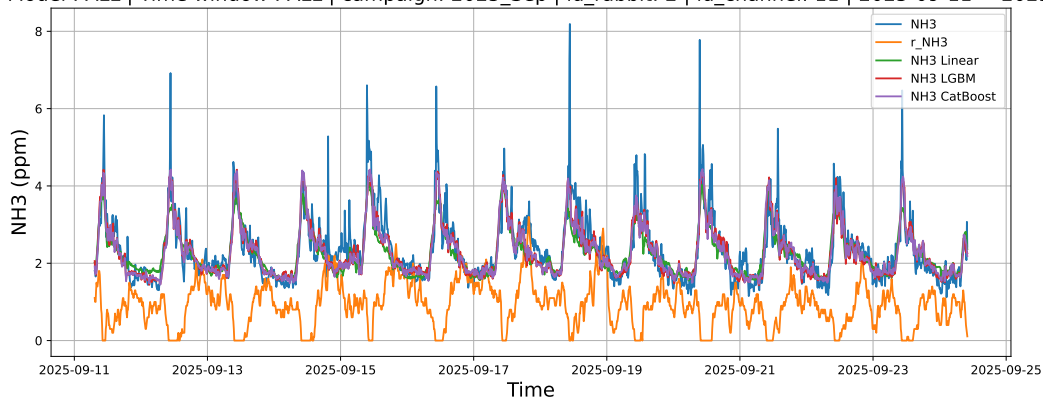
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

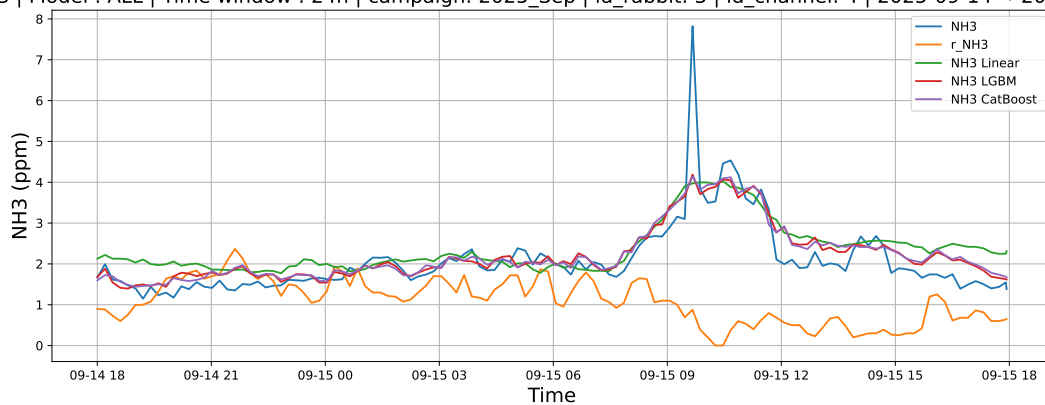
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

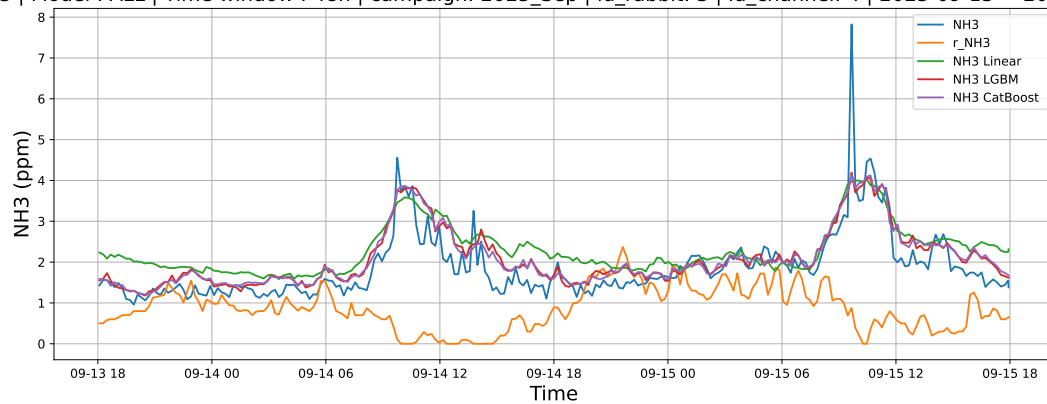
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



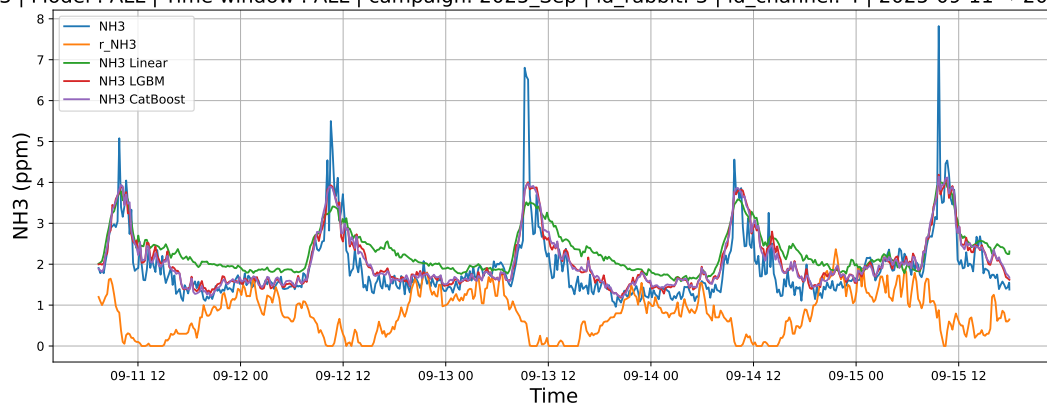
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

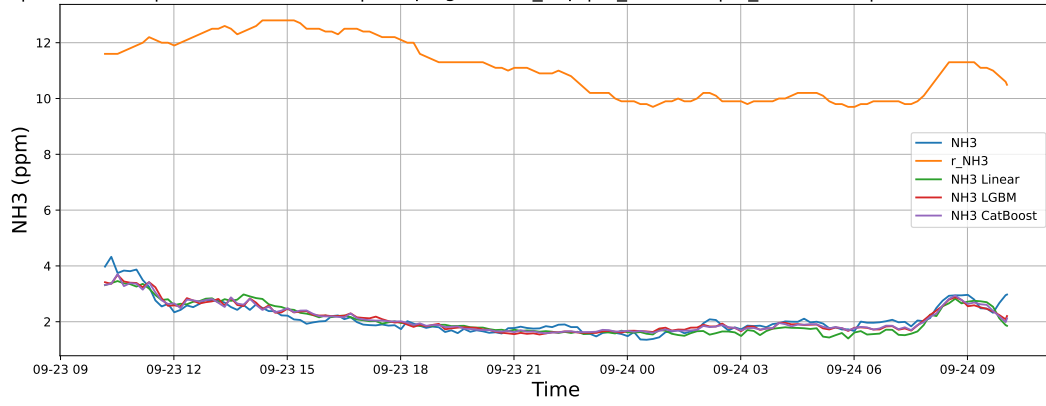
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

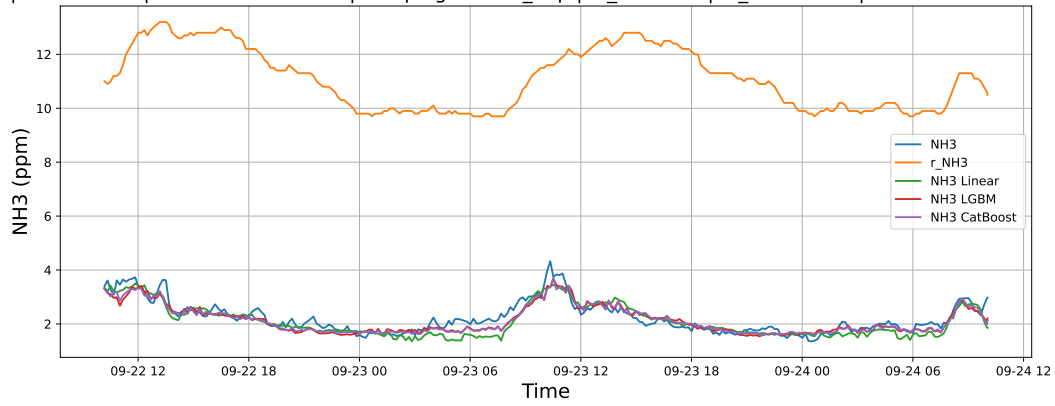
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



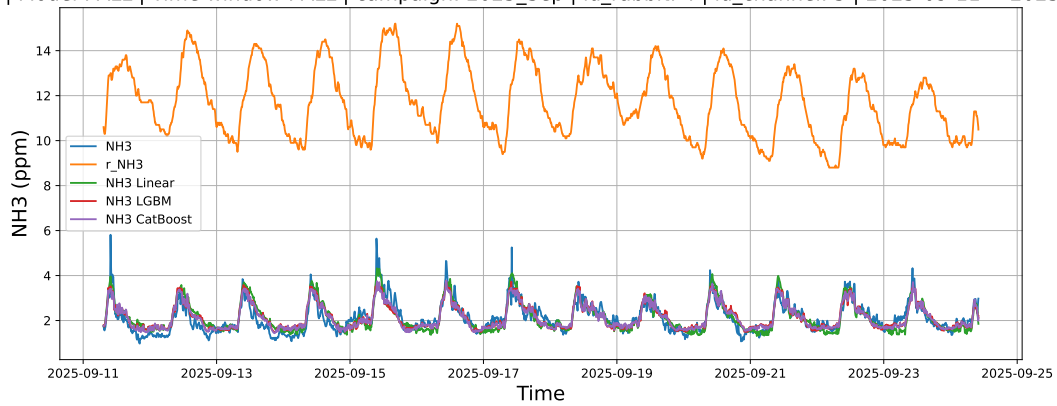
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

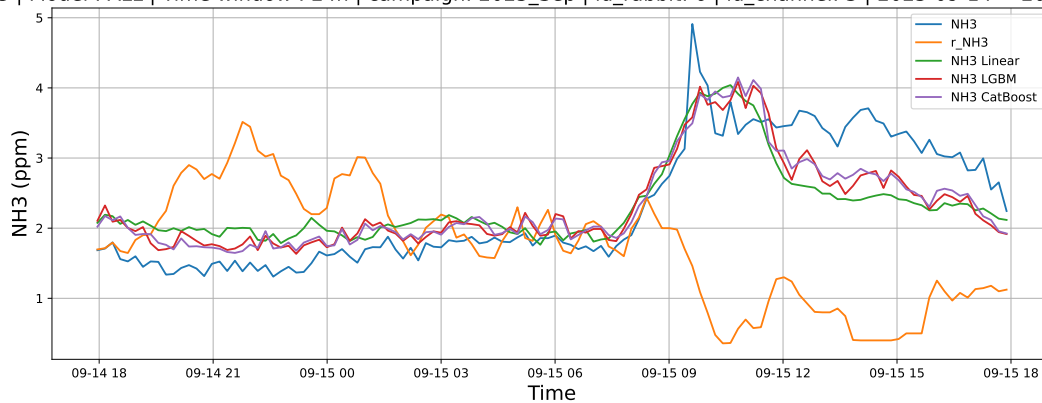
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

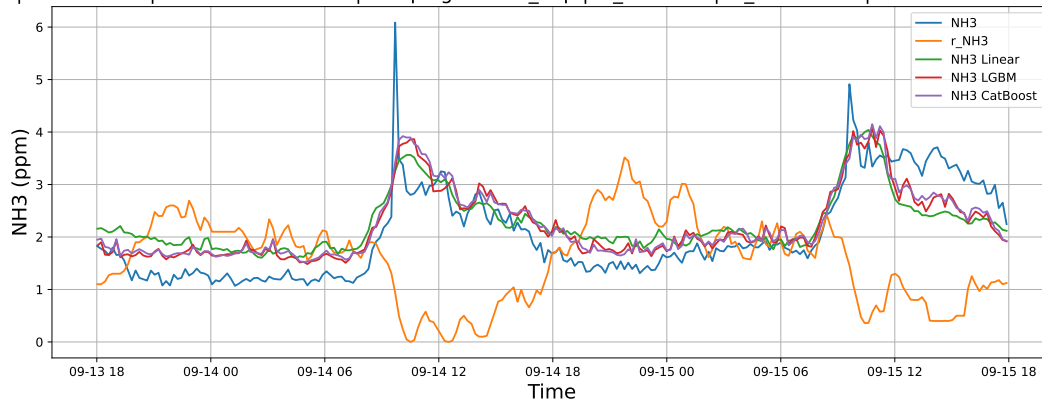
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

